

IMPERIAL

**Department of Mechanical Engineering
Advanced Mechanical Engineering Course**

**Analytical and Computational Investigation of Noise
Radiation and Propagation from Lifting Devices**

By

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Abstract

Open rotor aeroengines offer significant improvements in propulsion efficiency, but expose fan-generated noise directly to the atmosphere. Predicting this noise requires robust acoustic solvers. This report outlines the preliminary development and validation of an analytical Ffowcs Williams-Hawkings (FW-H) code using a permeable surface, retarded-time formulation. The numerical integration logic was evaluated against exact analytical derivations for both monopole and dipole sources under stationary, co-translating, and relative-motion conditions.

The code demonstrated good accuracy ($< 1\%$ error) for monopole sources, in which it captures convective terms and Doppler phase shifting, given sufficient grid and temporal resolutions. For dipole sources, derived analytical solutions were mathematically proven to fail when physical assumptions are violated. To bypass these limitations, a dual-monopole approximation was implemented and validated. This approach is sufficiently robust across all regimes, as it preserves the convective amplification factors that explicit Taylor expansions omit that was required when defining the dipole source term. While some validation work is still necessary, future work will move to a hybrid computational-analytical framework, using Unsteady Reynolds Averaged Navier Stokes (URANS)-FW-H hybrid to predict open rotor noise generation and propagation.

Acknowledgements

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Nomenclature

Variable	Definition
c	speed of sound
M	Mach number ($M = u_i/c$)
p	pressure
ρ	density
u	velocity
t	observer time
τ	retarded time
$\square^2$	d'Alembertian wave operator, $\square^2 = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$
∇^2	Laplacian differential operator, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
δ	Dirac delta function. Subchapter 2.5: infinitesimally small distance between two monopoles
δ_{ij}	Kronecker delta, $i = j, \delta_{ij} = 1; i \neq j, \delta_{ij} = 0$
y	static coordinate system
η	moving coordinate system
J	Jacobian of $y \rightarrow \eta$ coordinate change
$q(x, t)$	monopole source strength
$f_i(x, t)$	dipole source strength. Appendix C, arbitrary response.
T_{ij}	Lighthill stress Tensor, $T_{ij} = \rho u_i u_j + p_{ij} - c^2 p' \delta_{ij}$
λ	wavelength
f	wave frequency
$\hat{r}$	control surface radius
n	normal direction vector
ψ	Appendix C, arbitrary wave
m	Tyler-Sofrin spatial mode
n	Tyler-Sofrin temporal mode
θ	Azimuthal angle
ϕ	Phase angle
ω	Interaction frequency or source pulsation frequency, context are provided each time
Ω	Shaft speed

Variable	Definition (continued)
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B	Rotor blade count
V	Stator blade count
d	Circumferential distance between two grid points
N	Grid count

Subscripts	
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0	value in stationary flow
r	resolved in the direction of radiation
min	minimum/shortest value of interest
max	maximum/largest value of interest
$mono$	monopole source
$dipole$	dipole source
ref	reference value
num	numerical value
i	vector towards the i -direction
j	vector towards the j -direction. Subchapter 2.2 the j -th index of a sum
m	m-th Tyler-Sofrin spatial mode
n	n-th Tyler-Sofrin temporal mode
$fore$	Front blade row
aft	Aft blade row
s	Value at source, e.g., λ_s and f_s mean source wavelength and source frequency, respectively, prior to Doppler-shifting at the observer

Superscripts	
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'	perturbation component, e.g., $p' = p(x, t) - p_\infty$
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Chapter 1

Introduction

1.1 Motivation

Due to environmental concerns and the availability of fossil fuels, the efficiency of civil aero-engines has continuously been improved over the last few decades. Modern engines have increased the bypass ratio to improve efficiency [1], which is supported by increasing the nacelle and fan sizes. The open rotor (also known as "propfan") architecture improves on this by placing larger fans outside of the nacelle, potentially further reducing fuel consumption by 10-20% [2] while producing more thrust than existing engines and less carbon emissions.

However, aeroengine nacelles are designed to suppress the fan-generated noise through the use of acoustic linings, so this concept exposes it directly into the atmosphere, causing high noise pollution. An investigation of this noise generation and its propagation is required to develop suppression methods to reduce this noise. In this work, a hybrid approach between the Ffowcs Williams-Hawkings (FW-H) method and CFD will be used to predict the noise generation and propagation of an open rotor engine.

1.2 Open Rotor Noise Background

Unlike conventional turbofan engines, open rotor architectures lack a nacelle with acoustic liners to suppress the noise. Therefore, the noise generated by the rotors is radiated directly into the atmosphere without suppression. Open rotor noise is caused by the aerodynamic interactions and acoustic interference (inter-rotor and inter-mode interferences) [3]. The reason for this is the acoustic pressure fields generated by rotating blade rows and their aerodynamic interactions. To understand these modal interactions, the Tyler-Sofrin theory for rotor-stator wake interaction is used as the foundation.

The acoustics of a single rotor blade passing a stationary vane can be described as a periodic pressure fluctuation repeating at the blade-passage frequency. At a reference position, this fluctuation is expressed as a Fourier series in time, which can then be expanded into a

double Fourier series in space and time to describe the spatial-temporal components of the pressure field. For a single rotor blade-stator interaction, this component is defined as,

$$p'_{m,n} = a_{m,n} \cos(m\theta - nB\Omega t + \phi_{m,n}) \quad (1.1)$$

which represents a traveling wave with m lobes.

When this single-event model is superposed in an array of V evenly spaced stator vanes, the pressure contributions from each vane are shifted in space by the azimuthal angle $\Delta\theta = 2\pi/V$ and in time by $\Delta t = \Delta\theta/\Omega$. Most of these interferences are destructive; however, when $\Delta\theta = \Delta t$, the lobe number m is conserved to satisfy the Tyler-Sofrin mode,

$$m = nB + kV \quad (1.2)$$

where k is any positive or negative integer.

Extending this to the Contra-Rotating Open Rotor (CROR) architecture, this mechanism governs the inter-rotor interaction, with the kinematics being driven by two independently rotating blade rows. By using the aft blade as the reference frame, it acts as a stationary stator where $V = B_{aft}$ vanes spinning at an absolute angular velocity of Ω_{aft} . The front rotor, having B_{fore} blades, interacts with this aft row at a relative angular velocity of $\Omega_{fore} - \Omega_{aft}$. Applying the Tyler-Sofrin condition in this relative frame changes Equation 1.2 into $m = nB_{fore} + kB_{aft}$. Transforming this back to the absolute stationary frame shows the interaction frequency by adding the reference frame's rotation,

$$\omega = nB_{fore}\Omega_{fore} + kB_{aft}\Omega_{aft} \quad (1.3)$$

Consequently, the pattern spin speed of spatial mode m , evaluates to,

$$\Omega_m = \frac{\omega}{m} = \frac{nB_{fore}\Omega_{fore} + kB_{aft}\Omega_{aft}}{nB_{fore} + kB_{aft}} \quad (1.4)$$

and this kinematic relationship is the reason why CRORs generate such loud noises. As the rotors are rotating in opposite directions, Ω_{fore} and Ω_{aft} have opposite signs; therefore, certain harmonics will create combinations of n and k that will cause the denominator in Equation 1.4 to be very small. This increases the pattern spin speed Ω_m significantly. If the spinning mode's tip Mach number is subsonic, the sound intensity decays exponentially with distance from the rotor; however, for the case of CRORs, this spinning mode may go above supersonic, which propagates the sound unattenuated into the far-field [4].

Dunn, et al., [5] examined the noise data indicating that the frequencies corresponding to shaft orders dominate the sound field. Because the propagation of this noise is governed by

the blade counts and rotational speeds, the sound field is sensitive to the rotor's kinematic design. It is found that the dominating noise comes from shaft orders that correspond to the blade passing frequency (BPF) of $1 \times BPF_{aft}$, $1 \times BPF_{fore}$, $BPF_{fore} + BPF_{aft}$, $2 \times BPF_{aft} + 1 \times BPF_{fore}$, and $1 \times BPF_{aft} + 2 \times BPF_{fore}$, which confirms the inter-rotor and inter-mode interferences.

1.3 Hybrid Noise Prediction Background

Predicting open rotor noise requires solving the inhomogeneous wave equation. The FW-H equation provides the standard analytical framework for this process [6]. In this work, the permeable surface formulation (Formulation 1A) is utilised [7, 8, 9]. This formulation was chosen as it captures all noise sources, including quadrupoles (sounds from non-linearities such as shocks and turbulence), via surface integration, provided the control surface fully encapsulates the non-linearities [10], which avoids the computationally expensive volume integrals prohibitive to noise prediction.

In modelling acoustic wave propagation, a few methods are available such as Direct Noise Computation (DNC), FW-H analogy, and Hanson's model, to name a few. It is most practical, however, to conduct the analysis through a hybrid approach, in which flow fields surrounding the noise source are computed using CFD, and the far-field acoustic propagation are computed separately, such as using the FW-H method [11, 12]. This is to avoid the high computational cost of directly resolving the acoustic waves, while having a good geometric sensitivity of the sound source [13, 14, 15], unlike pure analytical approach. The schematic below in Figure 1.1 shows the available methods as well as highlights the current method used in this work.

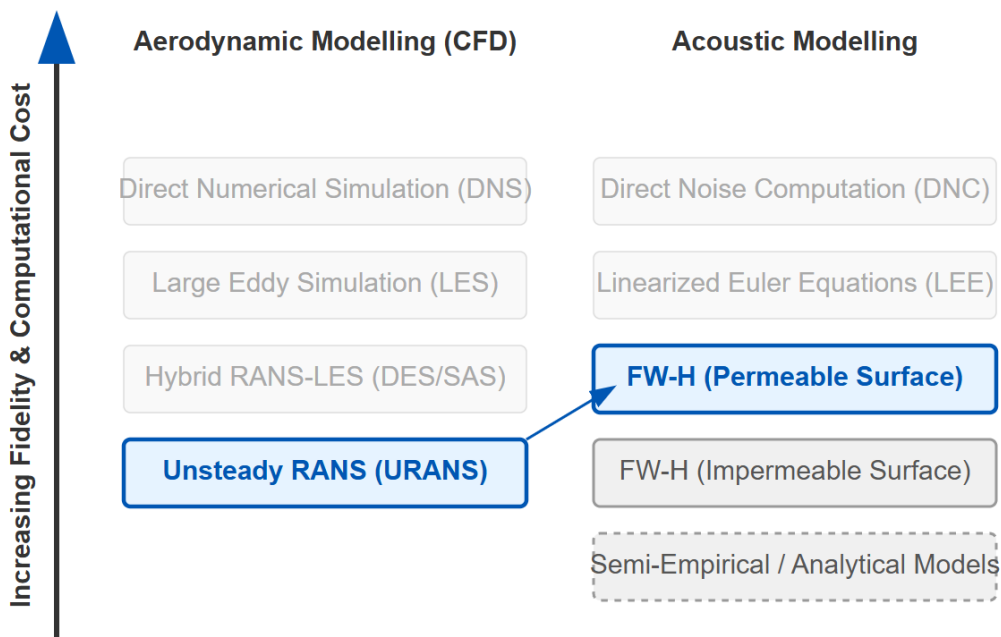


Figure 1.1: Hybrid modelling approach schematic

In this work, the FW-H method is validated and verified for its sensitivity using theoretical point sources, as illustrated in Figure 1.2, to build confidence in its capabilities; the detailed description of which will be provided in the next chapter. After confidence in the tool has been established, CFD will be used as an input to the FW-H code to form a hybrid method [16, 10, 17, 18], which replaces the point source in the illustration below with the CFD domain. As the interest in this work is to predict the tonal noise generated by the open rotor, the CFD framework used is the Unsteady Reynolds Averaged Navier-Stokes (URANS), as it is considered sufficient in providing a good balance between accuracy and computational cost for this purpose [14, 19]. Should a future interest arise in predicting broadband noise, other frameworks such as Detached Eddy Simulation (DES) [20] or Large Eddy Simulations (LES) can be implemented alongside the validated FW-H method.

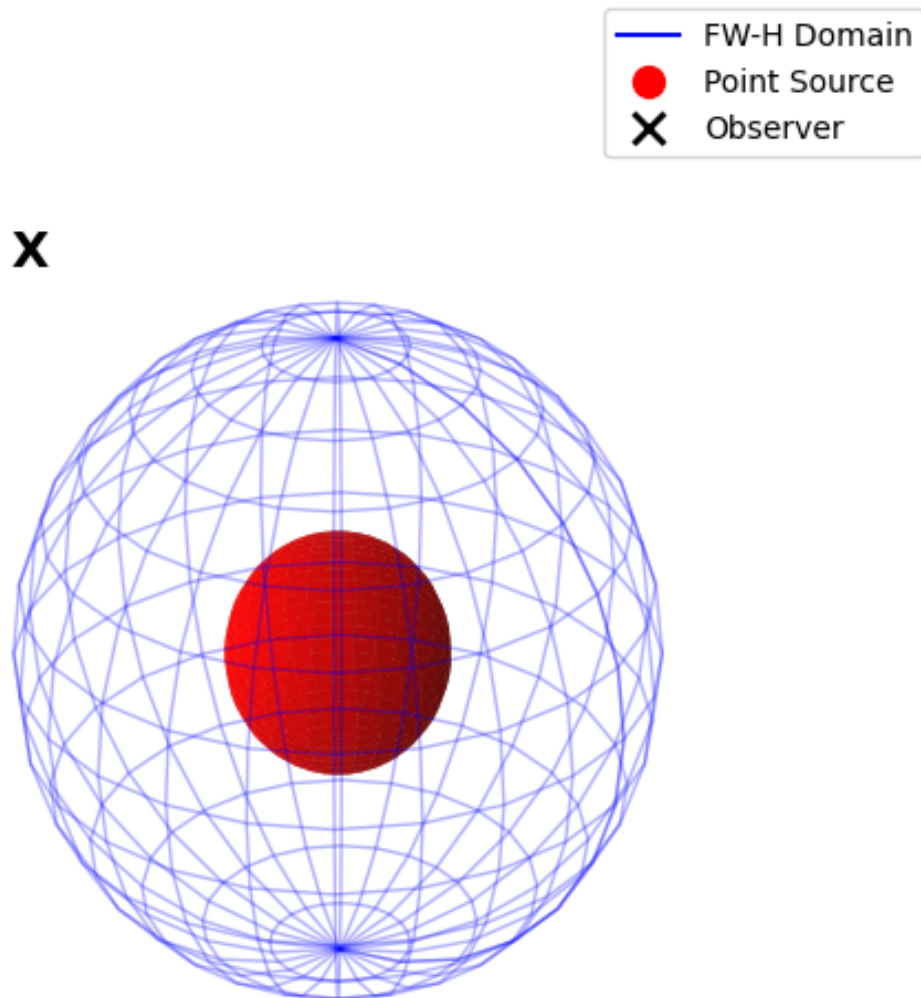


Figure 1.2: Hybrid domain illustration

1.4 Research Objectives and Outlines

The objectives of this work are:

1. **To develop an acoustics prediction method using the permeable surface form of the FW-H equation, then validating it using fundamental sound sources**

This part of the work is carried out in Chapter 2. The background of the permeable surface form of the Ffowcs Williams-Hawkings method is outlined, and the definition of the analytical thickness and loading (monopole and dipole, respectively) sound sources are also provided. These analytical sound sources are used as a baseline to compare the numerical FW-H results against, to build confidence in the code. The validation of the numerical FW-H method is performed using test cases that are described within the chapter.

2. **To develop an understanding of the sensitivity of the FW-H method to the temporal and spatial discretisation**

This is carried out in Chapter 3. Using monopole sound sources, the code is further tested for its sensitivity against the temporal and spatial discretisation. Discretisation parameters are introduced early in the chapter to describe their definitions and how they are used to parametrise the different tests. There are two main tests that are conducted; i.e., mimicking the sound from a known propeller using a ring of multiple stationary monopoles (also known as multipoles); and reproducing a rotating analytical sound source with varying Mach numbers and frequency ratios using a singular rotating monopole. These tests are devised to discover the code's sensitivity to the discretisation methods with multiple harmonics as well as Doppler-shifted wave properties at different Mach numbers.

3. **To use the FW-H method to predict open rotor noise using an appropriate CFD data**

This is done in Chapter 4. Something something Tyler-Sofrin modes something something using validating FWH and discretising based on the verification tests something something

Chapter 2

Development and Validation of the FW-H Method

This chapter describes the details of the FW-H method and the analytical frameworks used in its validation. The formulation used in the numerical FW-H method is described in Section 2.1 below along with the description of how the computer code works. The following sections will then describe the test cases used to validate the code; describe the analytical solutions of the monopole and dipole used to compare against the numerical solution; and finally, the validation data is presented to build confidence in the numerical code itself.

2.1 Description of the FW-H Method

2.1.1 Derivation of the FW-H Equation

To predict the open rotor noise, the FW-H equation in its permeable surface form provides a useful basis for noise prediction. This formulation is based on the governing laws of fluid motion, and the descriptions provided below are based on the original paper by Ffowcs Williams and Hawkings [7]. Consider a control surface S , which contains a combination of fluid and solid boundaries, while the region outside contains only fluid. The control surface S moves at a velocity v and is defined by the equation $f(x, t) = 0$, the following relationship then hold

$$\begin{aligned} f < 0, H(f) &= 0 \\ f > 0, H(f) &= 1 \\ \nabla_x f &= |\nabla_x f| \mathbf{n}, \mathbf{n} \text{ is the outward normal to } S \\ D_v f / Dt &= 0, \text{ so } \partial f / \partial t = -v \mathbf{n} |\nabla_x f| \\ \partial H(f) / \partial x_i &= \delta(f) \partial f / \partial x_i = \delta(f) \mathbf{n}_i |\nabla_x f| \end{aligned} \tag{2.1}$$

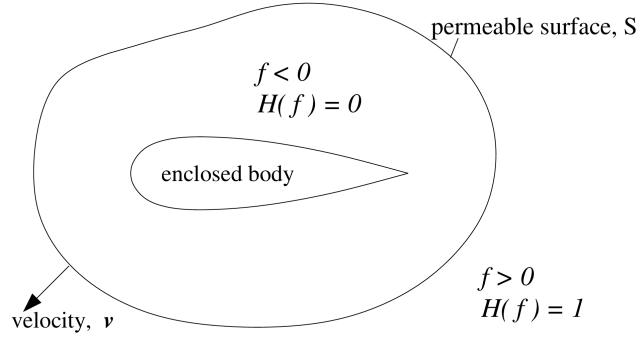


Figure 2.1: Permeable control surface

Generalised flow variables are denoted by an overbar. Outside of the surface, the variables are equal to the real flow variables, while inside they are assigned the value zero, therefore, there is a discontinuity across the surface S . Manipulating the mass and momentum conservation equations,

$$H(f) \left(\frac{\partial \rho'}{\partial t} + \frac{\partial \rho u_i}{\partial x_i} \right) = 0 \quad (2.2)$$

$$\frac{\partial \bar{\rho}'}{\partial t} + \frac{\partial (\bar{\rho} u_i)}{\partial x_i} = (\rho_0 u_{\mathbf{n}} + \rho' (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) |\nabla_x f|$$

$$H(f) \left(\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial}{\partial x_j} (p_{ij} + \rho u_i u_j) \right) = 0 \quad (2.3)$$

$$\frac{\partial (\bar{\rho} u_i)}{\partial t} + \frac{\partial}{\partial x_j} (\bar{p}_{ij} + \bar{\rho} u_i u_j) = (p_{ij} \mathbf{n}_j + \rho u_i (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) |\nabla_x f|$$

subtracting the divergence 2.3 from the time derivative 2.2 gives an inhomogeneous wave equation,

$$\square^2 p'(x, t) = \frac{\partial^2 \bar{T}_{ij}}{\partial x_i \partial x_j} - \frac{\partial}{\partial x_i} (p_{ij} \mathbf{n}_j + \rho u_i (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) |\nabla_x f| \quad (2.4)$$

$$+ \frac{\partial}{\partial t} (\rho_0 u_{\mathbf{n}} + \rho' (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) |\nabla_x f|$$

Which is the differential form of the FW-H equation, and is the governing equation for the generation and propagation of sound. Converting this differential form into an equivalent integral form is done using Green's function (see Appendix C). A coordinate change from the fixed one to a moving one with the control surface η is performed so the sources are at rest in the η space. The Jacobian for the coordinate change is J which represents the ratio of volume elements in the η and y spaces. Performing the steps above gives the inhomogeneous wave equation in the integral form of the FW-H equation,

$$\begin{aligned}
 p'(\mathbf{x}, t) = & \frac{\partial^2}{\partial x_i \partial x_j} \int_{-\infty}^{\infty} \frac{\overline{T_{ij}} \delta(g) J}{4\pi r} d^3\eta d\tau \\
 & - \frac{\partial}{\partial x_i} \int_{-\infty}^{\infty} \frac{(p_{ij} \mathbf{n}_j + \rho u_i (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) \delta(g) |\nabla_x f| J}{4\pi r} d^3\eta d\tau \\
 & + \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \frac{(\rho_0 u_{\mathbf{n}} + \rho' (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) \delta(g) |\nabla_x f| J}{4\pi r} d^3\eta d\tau
 \end{aligned} \tag{2.5}$$

The first right-hand side term, the quadrupole, is not dealt with explicitly [21, 3] as it is more significant for transonic and supersonic Mach numbers [7, 22]. By choosing a sufficiently large permeable surface, their contributions outside of this surface are negligible,

$$\begin{aligned}
 p'(\mathbf{x}, t) = & - \frac{\partial}{\partial x_i} \int_{-\infty}^{\infty} \frac{(p_{ij} \mathbf{n}_j + \rho u_i (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) \delta(g) |\nabla_x f| J}{4\pi r} d^3\eta d\tau \\
 & + \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \frac{(\rho_0 u_{\mathbf{n}} + \rho' (u_{\mathbf{n}} - v_{\mathbf{n}})) \delta(f) \delta(g) |\nabla_x f| J}{4\pi r} d^3\eta d\tau
 \end{aligned} \tag{2.6}$$

Using the notation by di Francescantonio [23] to simplify the algebra, new variables U_i and L_i are introduced to represent mass and momentum fluxes through the control surface,

$$U_i = (1 - \frac{\rho}{\rho_0})v_i + \frac{\rho}{\rho_0}u_i \quad L_i = p_{ij}\mathbf{n}_j + \rho u_i(u_{\mathbf{n}} - v_{\mathbf{n}})$$

Considering an undistorted control surface that translates at a constant and subsonic velocity, as described by Brentner and Farassat [24] is,

$$\begin{aligned}
 p'(\mathbf{x}, t) = & \frac{1}{4\pi} \int_S \left[\frac{\rho_0(\dot{U}_{\mathbf{n}} + U_{\mathbf{n}})}{r(1 - M_r)^2} \right]_{\tau^*} dS(\eta) \\
 & + \frac{1}{4\pi} \int_S \left[\frac{\rho_0(r(dM/d\tau)_r + c(M_r - |M|^2))}{r^2(1 - M_r)^3} \right]_{\tau^*} dS(\eta) \\
 & + \frac{1}{4\pi c} \int_S \left[\frac{\dot{L}_r}{r(1 - M_r)^2} \right]_{\tau^*} dS(\eta) \\
 & + \frac{1}{4\pi} \int_S \left[\frac{L_r - L_m}{r^2(1 - M_r)^2} \right]_{\tau^*} dS(\eta) \\
 & + \frac{1}{4\pi c} \int_S \left[\frac{L_r(r(dM/d\tau)_r + c(M_r - |M|^2))}{r^2(1 - M_r)^3} \right]_{\tau^*} dS(\eta)
 \end{aligned} \tag{2.7}$$

which is the numerical integration that is used by the numerical FW-H code to resolve the sound pressure at the observer locations. The first two terms describe the sound by monopole sources and the last three terms describe the sound by dipole sources.

2.1.2 Description of the FW-H Code

The FW-H code was written in Fortran 90 to implement the calculations computationally using equation 2.7. The program is summarised in the figure below,

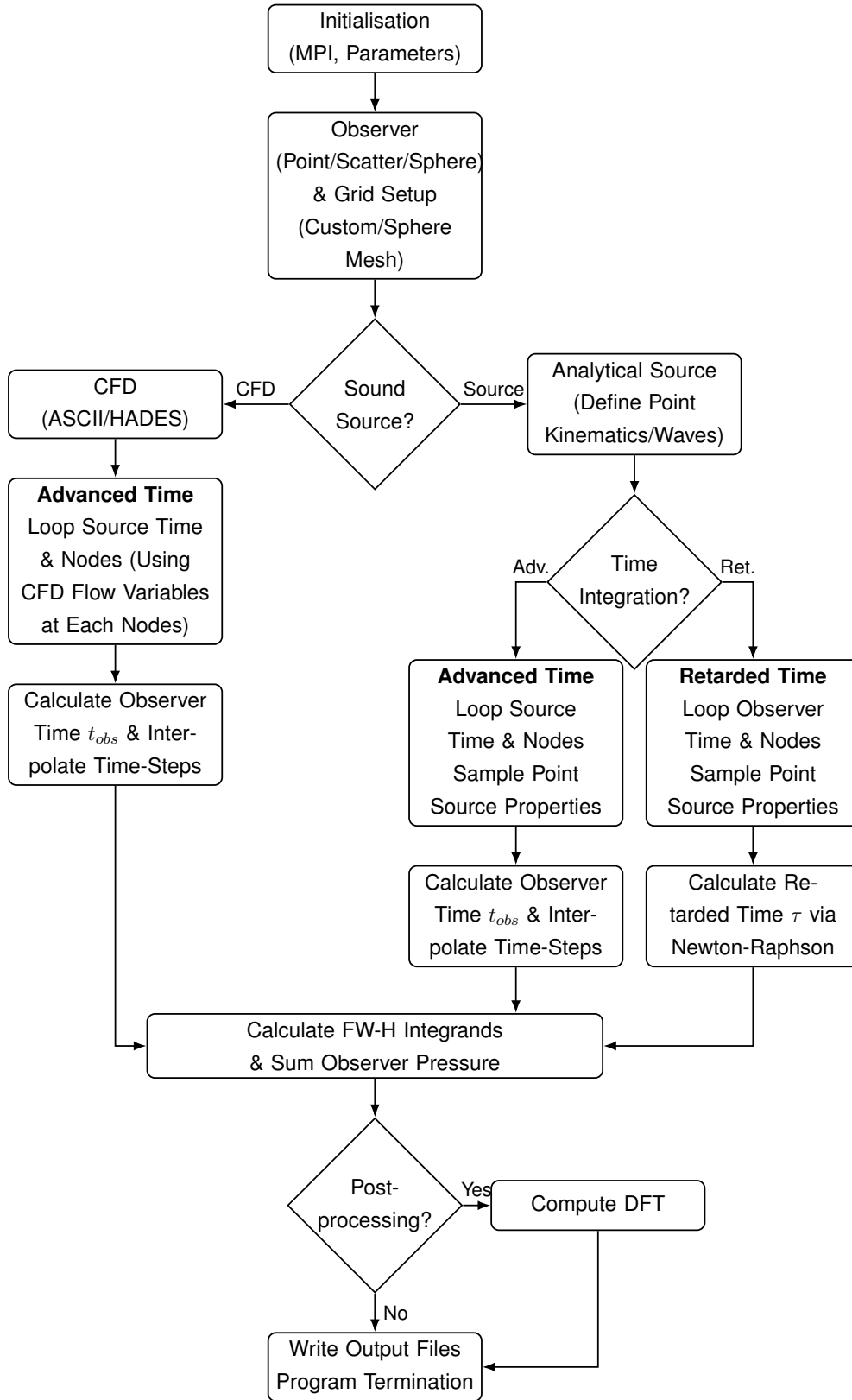


Figure 2.2: FW-H code flowchart: the code can receive both CFD data and analytical point sources as input, while using either advanced time or retarded time integration approaches

The ambient flow properties are first defined to be used as reference values. The code then checks whether the input sound source is an analytical point source, (simply referred to as, "Source" in the code) or discrete CFD input data (referred to as "ASCII/HADES"). The analytical point source will be described in the following sections and is used for the validation and sensitivity analyses conducted in this chapter and Chapter 3 as a reference value against which the numerical results are compared. CFD data is used in the Chapter 4 following the validation and sensitivity analysis of the FW-H code.

Following the source definition, the time integration approach is defined. Two time integration methodologies are implemented:

1. **The Retarded Time Method:** This approach is only available for the point sources. It iterates over a uniform grid of the observer's reception time, t , and implicitly calculates the corresponding source emission time τ . Because the distance between the source and observer, $r(\tau)$, is a function of an unknown emission time, the retarded time, $\tau = t - r(\tau)/c$ is resolved iteratively using a Newton-Raphson root-finding algorithm [25].
2. **The Advanced Time Method:** Also known as source-dominant formulation [26], this approach is compatible with both point sources and CFD inputs. As τ is known in this method, it instead iterates over the source's discrete emission times and computes the reception time at the observer $t = \tau + r(\tau)/c$. Because the motion of the sources may be non-uniform, the reception times are interpolated onto a uniform observer time grid.

Following the time integration, the surface integration in Equation 2.7 is resolved on the permeable surface grids to predict the observer acoustic pressure at time τ using a Riemann summation. This integration is performed at every time step to resolve the acoustic pressure time history at the observer location. Finally, the code can optionally post-process the time history data using a Discrete Fourier Transform (DFT).

It should be noted that as the surface integration and the time-marching use a finite number of discrete grids and time steps, respectively, both the validation and sensitivity studies will also evaluate how varying the spatial and temporal resolutions affects the numerical solution's accuracy. These are done to understand the code's behaviour in replicating the analytical solution, and therefore, to understand how to use it on a real CFD case.

When using the analytical source as input, the code also computes a direct analytical calculation, the descriptions of which are provided in Sections 2.3 and 2.5. These are calculated to give an analytical baseline against which the numerical FW-H results are compared. The following subchapters will discuss the validation of the FW-H code by comparing it against the known analytical solution.

2.2 Validation of the FW-H Code

To validate the numerical integration of the FW-H code, several test cases are developed where an analytical point source are used as an input. This section describes the validation methodology and the test cases. Figure 2.3 below illustrates the validation framework.

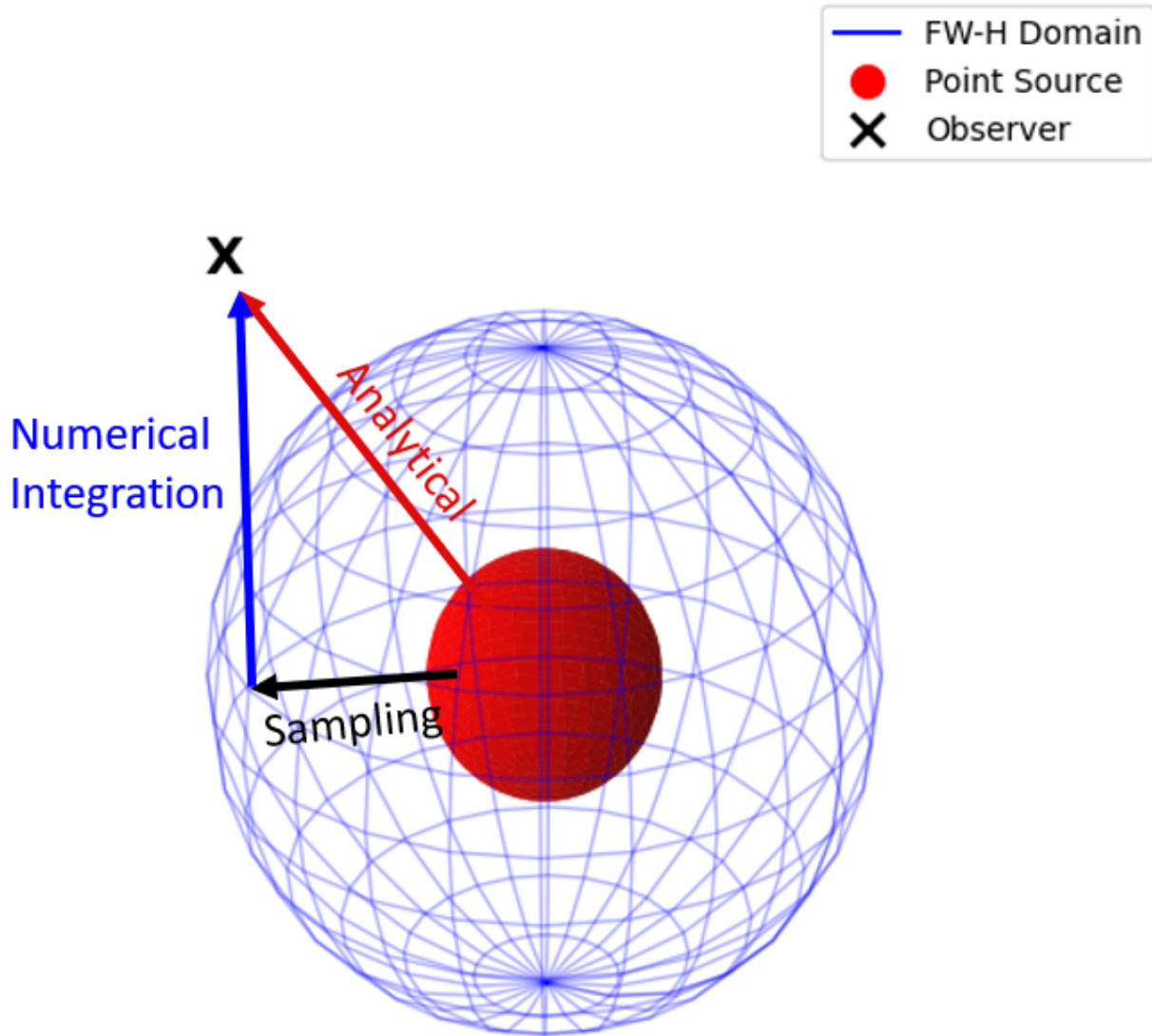


Figure 2.3: Validation framework

The numerical code samples the pressure generated by the source on the permeable boundary (black arrow), followed by surface integration of pressure to calculate the acoustic pressure at the observer (blue arrow). The analytical solution is calculated directly from the source to the observer (red arrow).

2.2.1 Test Cases

The following are the test cases used in the validation:

- Stationary and co-translating point source and observer. This test case is used as a baseline to test the overall prediction of the zero relative motion between point source and observer. The tests will also demonstrate the similarity between the two cases.

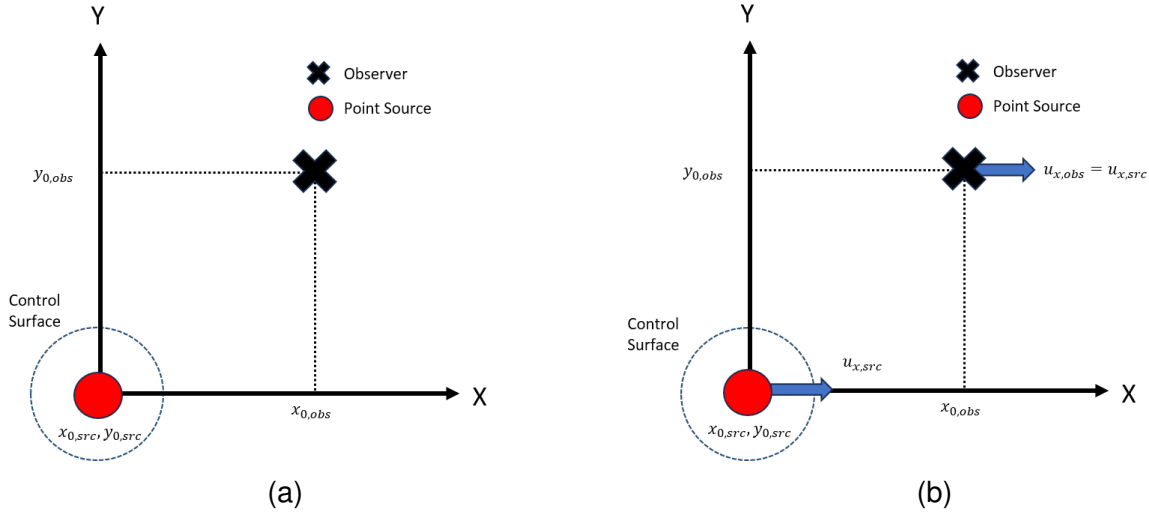


Figure 2.4: Illustration of, a) stationary point source and observer, and b) co-translating point source and observer cases

- Translating point source and stationary offset observer. This test case demonstrates the prediction with cases of non-zero relative motion between point source and observer. The observer is placed offset from the source's axis of motion such that the source can approach and move away from the observer in a single test case. This test case is done only for the monopole.

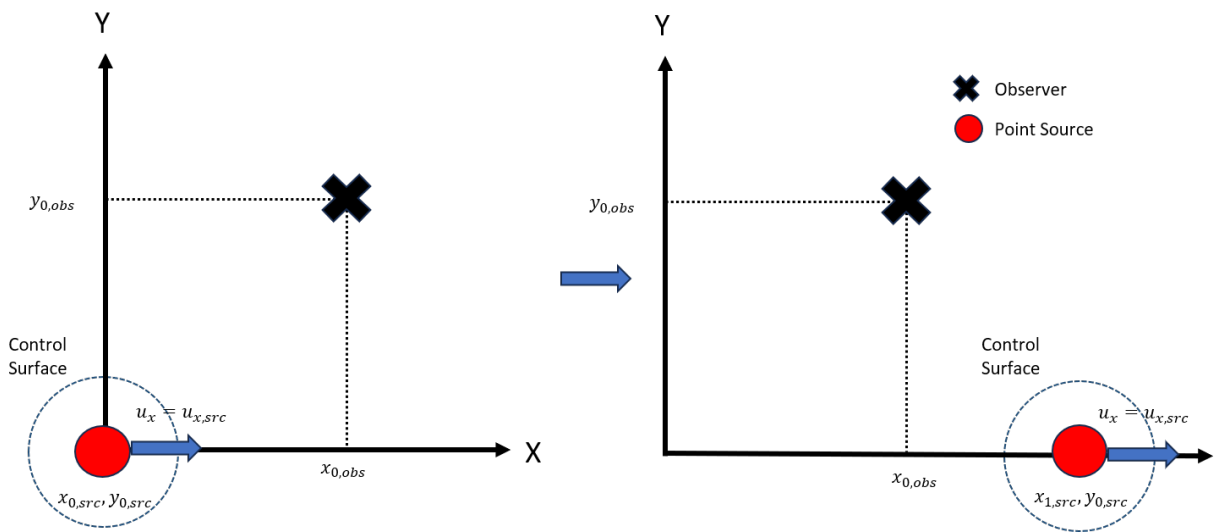


Figure 2.5: Illustration of translating point source and stationary observer cases

- Translating point source and stationary inline observer. This test case is done for the dipole test case to demonstrate the non-zero relative motion. The observer is placed inline instead of offset unlike the previous case to simplify the problem for the dipole's derivation in 1-D. The simulations are defined to finish before the control surface touches the observer to avoid singularity.

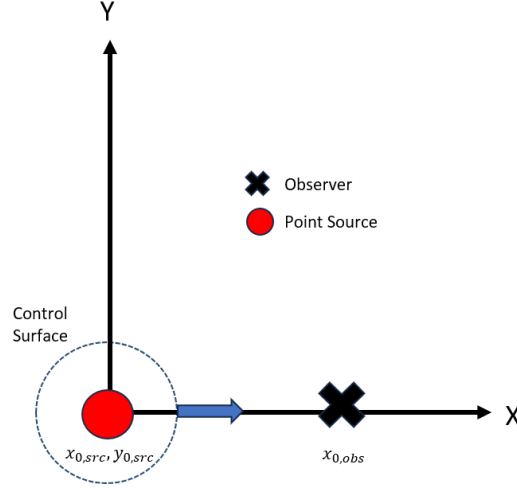


Figure 2.6: Illustration of translating point source and inline stationary observer case

2.2.2 Error Quantification

The errors of the numerical integration are quantified using the L2-norm error,

$$Error = \frac{\sum_j (p_{j,ref} - p_{j,num})^2}{\sum_j (p_{j,ref})^2} \quad (2.8)$$

which uses the same approach as Casalino's paper on the Advanced Time approach [26]. This approach is used to compare the instantaneous values of the numerical FW-H integration against the analytical model at every point in time; therefore, any deviations from the analytical results are evaluated. If the number of data points (i.e., number of time-steps) between the analytical and numerical solutions differ, such as during tests of varying temporal resolutions, the coarser numerical results are linearly interpolated to make up for the missing time-steps.

2.3 Analytical Solution for a Monopole

Analytical approaches for monopole sources have been validated in previous work by Char and Thomas [25, 17]. This section describes the derivation of the analytical monopole to provide a background on the analytical solution provided in the validation cases. Monopoles or thickness sources are sounds that are equivalent to mass supply, described as the following,

$$\square^2 p'(\mathbf{x}, t) = \frac{\partial}{\partial t} (Q(t) \delta(\mathbf{x} - \mathbf{y}(t))) \quad (2.9)$$

Convolving the above with the Green's function in 3D free-space (see Appendix C, described as,

$$G = \frac{\delta(g(\tau))}{4\pi r} \quad (2.10)$$

where $g(\tau) = t - \tau - \frac{r(\tau)}{c}$ and $r(\tau) = |\mathbf{x} - \mathbf{y}(\tau)|$, gives the following,

$$p'(\mathbf{x}, t) = \frac{\partial}{\partial t} \left[\frac{Q(\tau)}{4\pi r(1 - M_r)} \right]_{\tau^*} \quad (2.11)$$

which is the description of the monopole used in this work.

2.3.1 Monopole Pressure Equation

Taking the monopole definition as described in Equation 2.11, taking the differentiation using the definitions as described in Appendix B, the analytical solution for pressure is obtained,

$$p'(\mathbf{x}, t) = \frac{1}{4\pi} \left[\frac{\partial Q(\tau)/\partial \tau}{r(1 - M_r)^2} + \left(\frac{\partial M}{\partial \tau} \right)_r \frac{Q(\tau)}{r(1 - M_r)^3} + \frac{Q(\tau)c(M_r - |M|^2)}{r^2(1 - M_r)^3} \right]_{\tau^*} \quad (2.12)$$

The first term on the right-hand side describes the far-field propagation of the monopole due to its strength's temporal gradient, $\partial Q/\partial \tau$, the second term describes the far-field propagation due to acceleration, $\partial M/\partial \tau$, and the third term describes the near-field propagation.

2.3.2 Monopole Velocity Equation

Using the momentum equation,

$$\rho_0 \frac{\partial u_i}{\partial t} + \rho_0 \left(u_i \frac{\partial u_i}{\partial x_j} \right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right) \quad (2.13)$$

Linearising the equation and assuming no mean flow, substituting the monopole definition into the pressure term, and convolving with Green's function to replace $\partial/\partial x_i$ terms with

$\partial/\partial t$ (see Appendix D), the solution for velocity is obtained,

$$u'_i = \frac{1}{4\pi\rho_0} \left[\frac{(\partial Q(\tau)/\partial \tau)r_i}{cr^2(1-M_r)^2} + \frac{Q(\tau)r_i}{r^3(1-M_r)^3} + \left(\frac{dM}{d\tau} \right) \frac{Q(\tau)r_i}{cr^2(1-M_r)^3} - \frac{Q(\tau)M^2r_i}{r^3(1-M_r)^3} - \frac{Q(\tau)M_i}{r^2(1-M_r)^2} \right]_{\tau^*} \quad (2.14)$$

where the first and third terms correspond to the far-field velocity perturbation components due to the strength and Mach number temporal gradients, respectively; and the second, fourth, and fifth terms correspond to the near-field perturbation components.

2.3.3 Monopole Density Equation

The expression for density is based on the assumption of a linear, isentropic flow. Therefore, the expression for density is,

$$\rho' = \frac{p'}{c^2} \quad (2.15)$$

which is valid for both monopole and dipole sources.

2.4 Validation of the FW-H Method for Monopole Sources

The point source in each case is prescribed with source strength of $Q(\tau) = \sin(10\tau)$, and is placed at the origin. Speed of sound, c , and the freestream air density ρ_0 are set to 340m/s and 1.2kg/m^3 , respectively. The Retarded and Advanced Time algorithms are both tested.

Three cases are defined to validate the monopole; the co-stationary, co-translating, and the relative motion cases. These cases provide an understanding of the code's ability in predicting simple sounds. The observer is initially located at $[10, 10, 0]$ for the co-stationary and co-translating cases, and at $[150, 10, 0]$ for the relative motion case. In the translating cases, the relevant points are moving at Mach 0.5 along the x-direction. The test cases are summarised in Table 2.1 below.

Table 2.1: Summary of Monopole Validation Test Cases

Case Name	Time Algorithm	M_{source}	$M_{observer}$	$x_{0,observer}$
Co-Stationary	Retarded & Advanced	[0, 0, 0]	[0, 0, 0]	[10, 10, 0]
Co-Translating	Retarded	[0.5, 0, 0]	[0.5, 0, 0]	[10, 10, 0]
Translating	Retarded	[0.5, 0, 0]	[0, 0, 0]	[150, 10, 0]
Translating	Advanced	[0, 0, 0]	[0.5, 0, 0]	[150, 10, 0]

The radius of the control surface is set to $5m$ and the surface is discretised to capture the pressure wavelengths at 275, 100, and 35 grid points per wavelength, which correspond to 40×40 , 15×15 , and 5×5 grid resolutions, respectively. Similarly, the temporal discretisation is set to capture frequencies at 30, 15, and 5 time steps per wave period.

2.4.1 Co-Stationary Monopole Case

Starting with the co-stationary case using the Retarded Time algorithm, it can be seen from Figure 2.7 below that the numerical FW-H method agrees well with the analytical solution; provided that the discretisation resolutions resolve the pressure waves radiated by the point source sufficiently. An under-prediction of the amplitudes are observed when using coarser grid resolutions, while using coarser temporal resolutions causes the FW-H prediction to lag behind the analytical solutions.

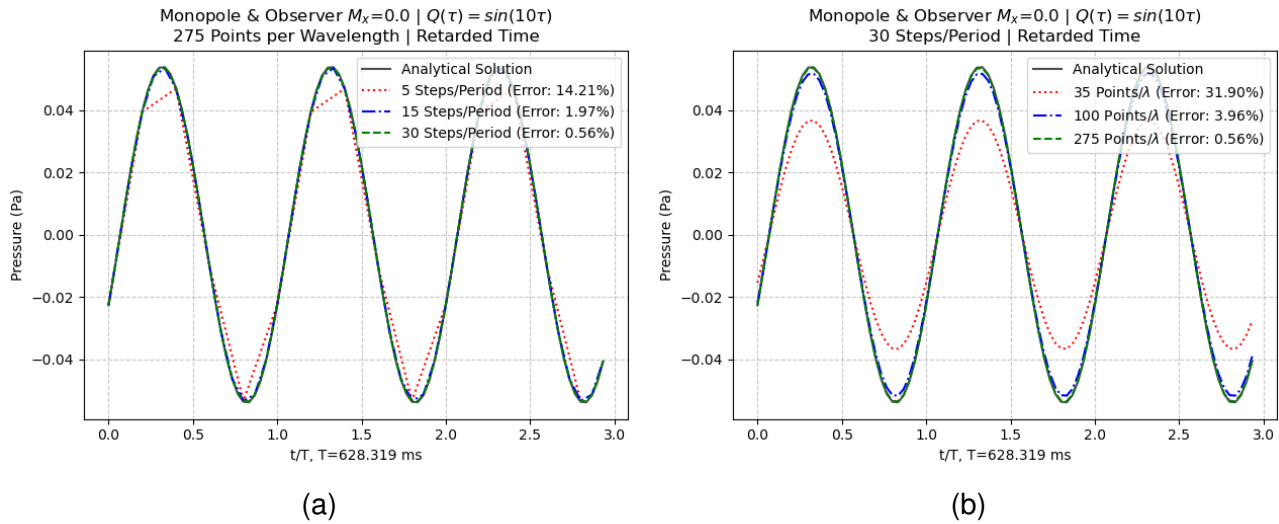


Figure 2.7: Co-stationary monopole case using the retarded time algorithm, varying the a) time-step sizing, and b) grid sizing

Regarding the amplitude under-prediction, it cannot be attributed to the code's inability to predict the wavelengths; as the wavelength in this case is ~ 213.6 meters, while the coarsest 5×5 grid resolution resolves this to 35 points per wavelength. Instead, this is largely a geometric error caused by the coarser grids that is approximating the control surface sphere as a polygon. On the other hand, the coarsest temporal resolution in this case was indeed unable to predict the waves well enough due to the insufficient temporal resolution.

Moving on to the Advanced Time algorithm, it should be noted that due to the nature of this algorithm, the sound signal received at the observer requires some post-processing work. This post-processing removes the first and last period of the observer signal which reads as zero. This post-processing was done prior to any error calculations or plotting; and therefore, all the results presented here have been cleaned up.

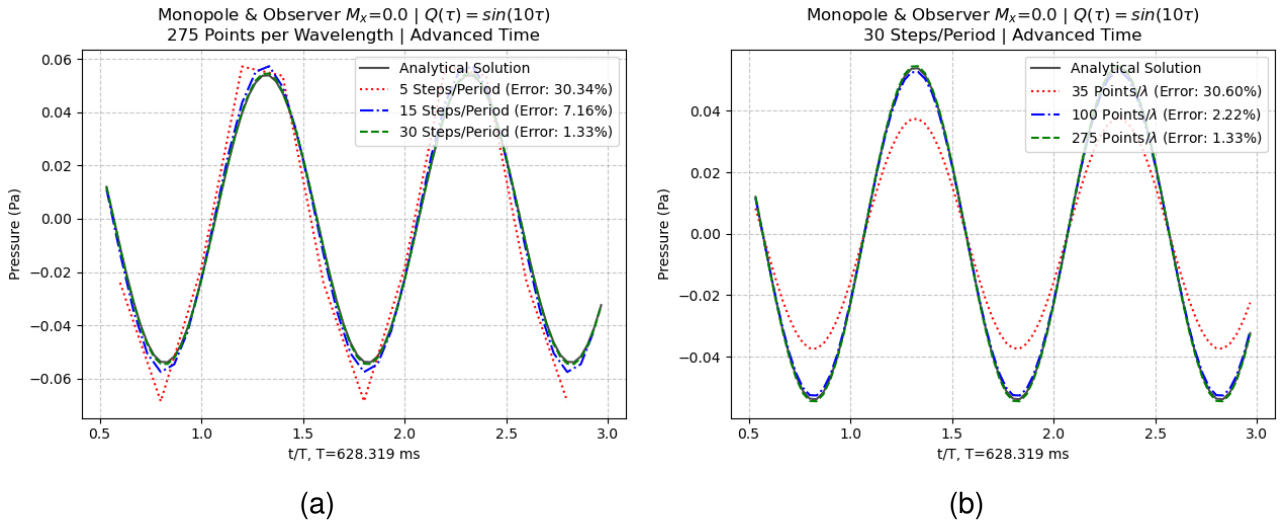


Figure 2.8: Co-stationary monopole case using the advanced time algorithm, varying the a) time-step sizing, and b) grid sizing

As with the Retarded Time results, the numerical integral using the Advanced Time method generally has a good agreement with the analytical solution. The errors are relatively higher with this method due to the time interpolation at the observer points; an inherent feature of the Advanced Time approach as described in Subsection 2.1.2. The effects of discretisation resolutions are also relatively similar to the Retarded Time solutions, where coarser time-step sizes somewhat lags behind the analytical solution, while the coarser grids under-predict the amplitude due to geometric error.

2.4.2 Co-Translating Monopole Case

The co-translating case only uses the Retarded Time algorithm as its purpose is to demonstrate how the signal is received at the observer when compared to the co-stationary case.

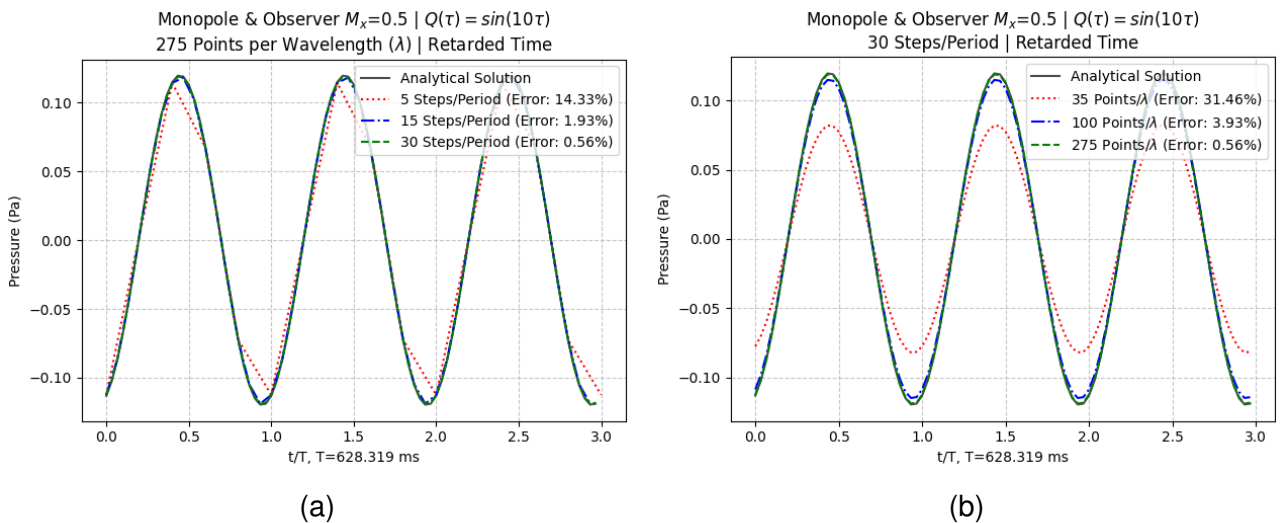


Figure 2.9: Co-translating monopole case using the retarded time algorithm, varying the a) time-step sizing, and b) grid sizing

Comparing these results with the co-stationary cases (Figure 2.7) shows a distinct phase difference and amplitude, despite both scenario having zero relative Mach number and the distance between the source and observer is kept the same. In the co-translating case, the source moves through the quiescent fluid medium, which causes a phase shift of $\sim \pi/2$ and an amplitude increase due to the convective term. Furthermore, the emission distance r in the retarded time, $\tau = t - r/c$, differs from the geometric distance because both points are moving through the medium during the propagation time. In terms of discretisation resolutions, the errors between this co-translating case and the retarded time co-stationary case are roughly the same, which suggest that despite there being some Doppler shifting due to the motion of the source, the observer's equal motion cancels out the errors which may arise.

2.4.3 Translating Monopole Case

For the translation case, the set up for the two time algorithms are slightly different, as summarised in Table 2.1. In the Retarded Time case, the source moves along the X-axis to the positive direction, while the observer remains at its initial position; while in the Advanced Time case, the source remains stationary while the observer moves parallel to the X-axis to the negative direction.

One reason for this differentiated approach is, at the time of the writing of this work, the Advanced Time algorithm does not support the permeable surface's tracking of the source's motion. This means, while the source can move within the permeable surface, the surface can only remain stationary in space; which is also the reason why the co-translating case was only tested using the Retarded Time algorithm. Another reason for this differentiated approach is to demonstrate the signal amplification due to Doppler shifts, which will be elaborated in the Advanced Time results.

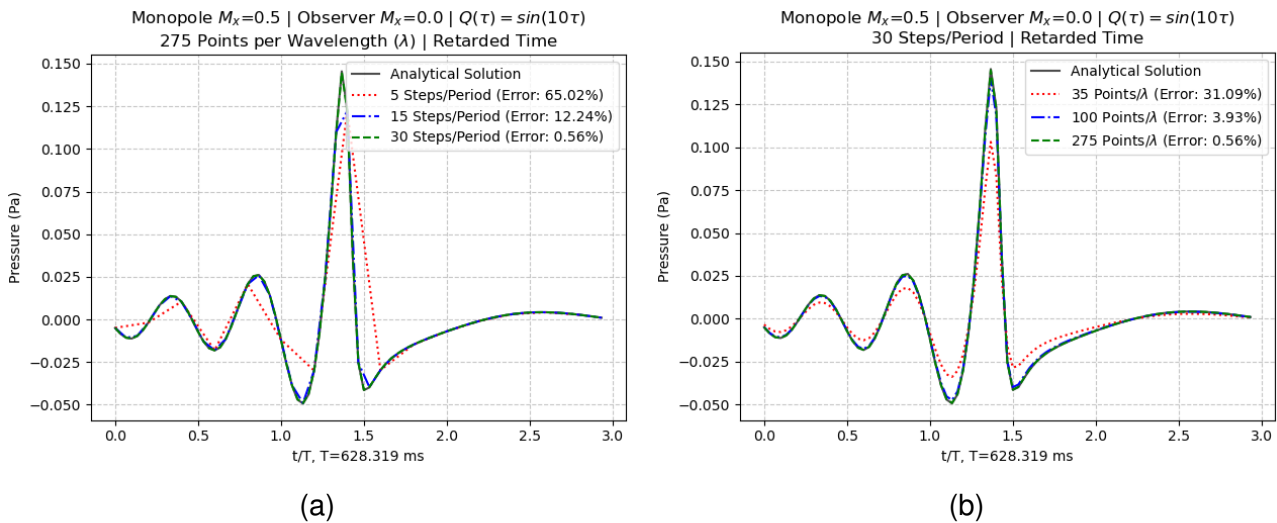


Figure 2.10: Translating monopole with stationary observer test case using the Retarded Time method, a) with varied time-step sizing, and b) with varied grid sizing

This case demonstrates the relative motion between the source and observer by imposing a relative Mach number between the two points. In Figure 2.10 the Doppler effect causes the perceived frequency and pressure amplitude at the observer to increase as the source approaches ($0 \leq t/T \leq 1.5$), and decrease as it moves away ($t/T > 1.5$). This wave compression and expansion effect of Doppler shift is a common occurrence in aeroengines, whereby as the blade rotates, its motion causes the blade to approach and move away from a stationary observer. As the Figure 2.10 above shows, the Retarded Time algorithm can predict this Doppler shift effects with sufficient discretisation resolutions.

Moving to the Advanced Time algorithm, when compared to the Retarded Time algorithm, one noticeable feature of the time-history signal of the former is the lack of the large amplitude spike as the observer moves towards the source.

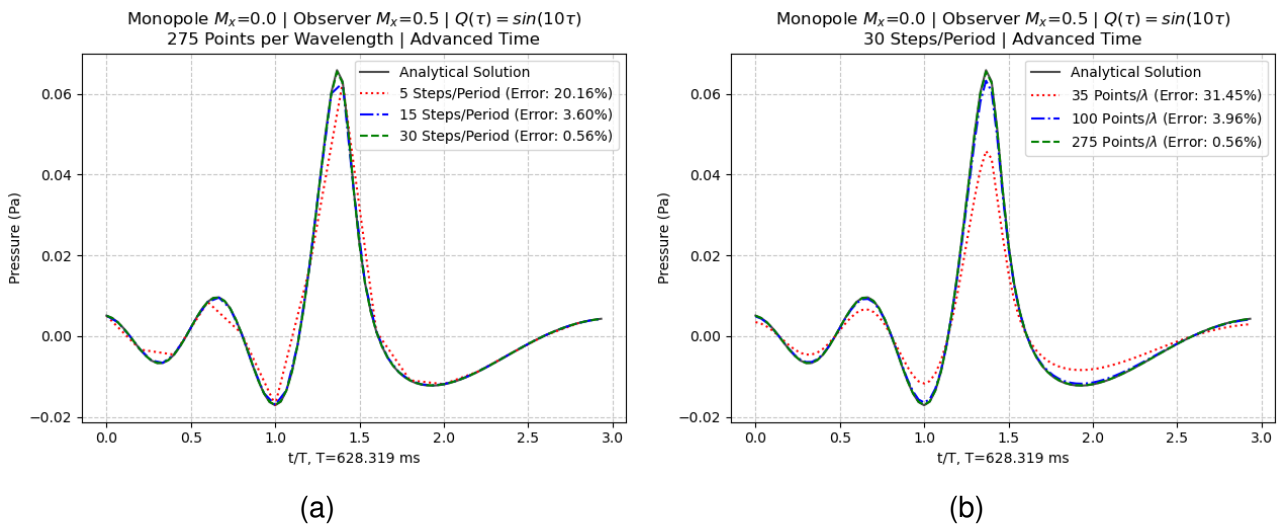


Figure 2.11: Translating observer with stationary monopole test case using the Advanced Time method, a) with varied time-step sizing, and b) with varied grid sizing

As can be seen from Figure 2.11, the frequency shifts as the observer is moving, but the pressure amplification peak only reaches $\sim 0.07Pa$, instead of $\sim 0.150Pa$ as in the Retarded Time case. This is not an issue with the Advanced Time algorithm itself, but in how the case was set up. This pressure amplification by a factor of $1/(1 - M_r)$ is most efficient in the direction of the source's motion, and gradually reduces to $1/(1 + M_r)$ in the opposing direction of the motion. However, it can be said from this result that the Advanced Time algorithm can be used to predict this Doppler shifting effect with a comparable accuracy to the Retarded time algorithm.

With these cases explored, it can be concluded that the FW-H code can sufficiently predict the sound signal coming from a thickness (monopole) noise. The next sections within this chapter will focus on the loading (dipole) noise sources, which is highly relevant the noise generated from an open rotor engine.

2.5 Analytical Solution for a Dipole

Dipoles or loading sources are sounds due to forces acting on the fluid, equivalent to body forces and momentum supply. It is also equivalent to two equal monopoles with opposite signs at an infinitesimally small distance. The dipole is described as the following,

$$p'(\mathbf{x}, t) = -\frac{\partial}{\partial x_i} \left[\frac{f_i(\tau)}{4\pi r(1 - M_r)} \right]_{\tau*} \quad (2.16)$$

In the dipole description above, the source forcing term f_i is not an arbitrary vector. To analytically model a dipole, the mathematical equivalence between monopoles and dipoles must be established. Sound pressure generated by a monopole source is defined as,

$$p'_{mono}(\mathbf{x}, t) = \frac{\partial}{\partial t} \left(\frac{Q(\tau)}{4\pi r|1 - M_r|} \right)$$

By describing a dipole as two monopoles of equal strength and opposite signs separated by an infinitely small distance δ , the dipole pressure is a superposition of the two monopoles,

$$p'_{dipole} = p'_{mono}(\mathbf{x}, t) - p'_{mono}(\mathbf{x} - \delta, t)$$

As δ tends to zero, this distance can be expressed as a first-order Taylor series expansion,

$$p'_{dipole}(\mathbf{x}, t) \approx -\delta \frac{\partial}{\partial x_i} \frac{\partial}{\partial t} \left[\frac{Q(\tau)}{4\pi r|1 - M_r|} \right]$$

Hence, the dipole source strength $f_i(t)$ can be defined as,

$$f_i(t) = 4\pi r|1 - M_r| \delta \frac{\partial}{\partial t} \left(\frac{Q(t)}{4\pi r|1 - M_r|} \right) \quad (2.17)$$

by defining the forcing term in this term and evaluating it using central differencing, the analytical code captures the expansion of this forcing term without requiring an analytical expansion.

2.5.1 Dipole Pressure Equation

Using the definition described in Equation 2.16, replacing the $\partial/\partial x_i$ term as outlined in Appendix D, and using the steps from monopole pressure, the pressure for an analytical dipole is obtained,

$$p'(\mathbf{x}, t) = \frac{(\partial f_i(\tau)/\partial \tau)r_i}{4\pi r^2 c(1 - M_r)^2} + \frac{f_i(\tau)r_i}{4\pi r^3 |1 - M_r|^2} - \frac{f_i(\tau)M_r}{4\pi r^2 |1 - M_r|^2} + \left(\frac{\partial M}{\partial \tau}\right) \frac{f_i(\tau)r_i}{4\pi r^2 c |1 - M_r|^3} + \frac{f_i(\tau)r_i(|M|^2 - |M_r|^2)}{4\pi r^3 |1 - M_r|^3} \quad (2.18)$$

where the first and fourth terms on the right-hand side are the far-field propagation due to changes in the source forcing term and Mach number, respectively; while the second, third, and fifth terms are the near-field propagation terms. The far-field terms are the primary reason for the noise propagation in rotors such as helicopter blades, propellers, and open rotor engines; as the blades rotate and push the air surrounding it, the local forcing strength and Mach number changes with time.

2.5.2 Dipole Velocity Equation

To analytically derive the velocity for a dipole, explicitly integrating the momentum equation like with the monopole is unviable, as this approach will return too many terms. Instead, limited solutions are obtained by applying some physical assumptions. The velocity derivations assume either a zero relative Mach number or a plane wave condition. An alternative method that uses the superposition of two equal monopoles of opposite signs will also be presented.

Dipole Velocity Equation: Zero Relative Mach Number Assumption

Assuming inviscid and no mean flow in the momentum equation, the pressure gradient is,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial^2}{\partial x_i \partial x_j} \left[\frac{f_i(\tau)}{4\pi r(1 - M_r)} \right]_{\tau^*}$$

Hence, the velocity is defined as,

$$u'_i = \frac{1}{4\pi\rho_0} \frac{\partial^2}{\partial x_i \partial x_j} \int_{-\infty}^t \frac{f_i(\tau)}{r(1 - M_r)} dt$$

substituting the integrand to use the time at the observer, τ , and assuming zero relative velocity; therefore, $dr/dt = 0$ and $M_r = 0$. Then, let $F_i(\tau) = \int f_i(\tau) d\tau \implies f_i(\tau) = \dot{F}_i(\tau)$, the analytical solution for dipole velocity that is valid for cases with zero relative Mach number is obtained,

$$u'_i = \frac{1}{4\pi\rho_0} \left(\frac{F_j(3\mathbf{n}_i\mathbf{n}_j - \delta_{ij})}{r^3} + \frac{f_j(3\mathbf{n}_i\mathbf{n}_j - \delta_{ij})}{cr^2} + \frac{\dot{f}_j(\mathbf{n}_i\mathbf{n}_j)}{c^2 r} \right) \quad (2.19)$$

where the first, second, and third terms on the right hand side are the near-, intermediate-, and far-field velocity propagation, respectively.

Dipole Velocity Equation: Plane Wave Assumption

The plane wave assumption allows relative Mach number between the source and observer, provided the control surface is within the far-field of the source. In the far-field, the pressure and velocity field perturbation components are in-phase, which is the reason of this requirement. The far-field is defined as,

$$k\hat{r} \gg 1 \quad (2.20)$$

Physically, this means the acoustic wavelength must be smaller than the control surface diameter, allowing the entire wavelength to be fully captured by the permeable sphere. Using the derivations in Appendix G, the velocity term with the plane wave assumption is obtained,

$$u'_i = \frac{\mathbf{n}_i}{\rho_0 c} p'(\mathbf{x}, t) \quad (2.21)$$

hence showing that the velocity perturbation component is indeed in-phase with the pressure.

Dual Monopole

An alternative approach to validate dipole sources is the dual monopole method. This approach entirely avoids the assumptions that were made in the previous subsections and instead uses two monopoles in the domain. This bypasses the case limitations imposed by the assumptions; therefore, when using this method on cases that would violate assumptions made above, this approach will remain accurate. However, to ensure that the dual monopole approach itself is valid, it needs to be compared against some known analytical solution, however limited the predictions from these solutions are.

Using the definition of a dipole, this method uses two equal monopoles of opposing signs, separated by a small distance δ , which in this case is defined as,

$$k\delta \ll 1 \quad (2.22)$$

To simplify the validation process, it is assumed in the test cases that the distance between the two monopoles is $\delta = 0.01/k$, which should still validate this assumption. Another advantage of this approach is that it allows the prediction of the directivity patterns of the dipole, the results of which will also be presented in the next section.

2.6 Validation of the FW-H Method for Dipole Sources

With the FW-H numerical solver validated for monopoles, the validation is extended to dipole sources. As mentioned in Subchapter 2.5, explicitly deriving the velocity terms for dipole sources is unviable, whcih led to the two approaches being tested separately to test the validity of each assumption and tests for cases on which these assumption are violated. The dual monopole approach will then redo all these cases to ensure its own validity. Table 2.2 below summarises the test cases for the dipole; the column "Radius" refers to the permeable surface radius.

Table 2.2: Summary of Dipole Validation Test Cases

No	Assumption	Radius	M_{source}	$M_{observer}$	$x_{0,observer}$	Validity
1	Zero Relative Motion	1.0m	[0,0,0]	[0,0,0]	[10,0,0]	Valid
2		1.0m	[0.5,0,0]	[0.5,0,0]	[10,0,0]	Valid
3		1.0m	[0.5,0,0]	[0,0,0]	[10,0,0]	Invalid
4	Plane Wave	500.0m	[0.5,0,0]	[0,0,0]	[500,0,0]	Valid
5		1.0m	[0.5,0,0]	[0,0,0]	[1500,0,0]	Invalid
6	Dual Monopole	1.0m	[0,0,0]	[0,0,0]	[10,0,0]	-
7		1.0m	[0.5,0,0]	[0.5,0,0]	[10,0,0]	-
8		1.0m	[0.5,0,0]	[0,0,0]	[10,0,0]	-
9		500.0m	[0.5,0,0]	[0,0,0]	[1500,0,0]	-
10		1.0m	[0.5,0,0]	[0,0,0]	[1500,0,0]	-

In the the following tests, only the Retarded Time algorithm will be used to allow the control surface to move along with the source. Only the finest discretisation resolution from the monopole tests will be used (275 grid points per wavelength and 30 time steps per period), to focus on validating the assumptions. Like the monopole cases, the source is set to radiate with the strength $Q(\tau) = \sin(10\tau)$ and is initially placed at the origin.

2.6.1 Dipole Derivation Validations

Unlike the monopole validation, the co-translating test case is omitted for the dipole cases to isolate the effects of a non-zero relative mach number. Therefore, the observer is stationary in all test cases. For all subsequent dipole test cases, the dipole axis and translation velocity are strictly imposed along the x -axis, while the observer maintains a transverse offset.

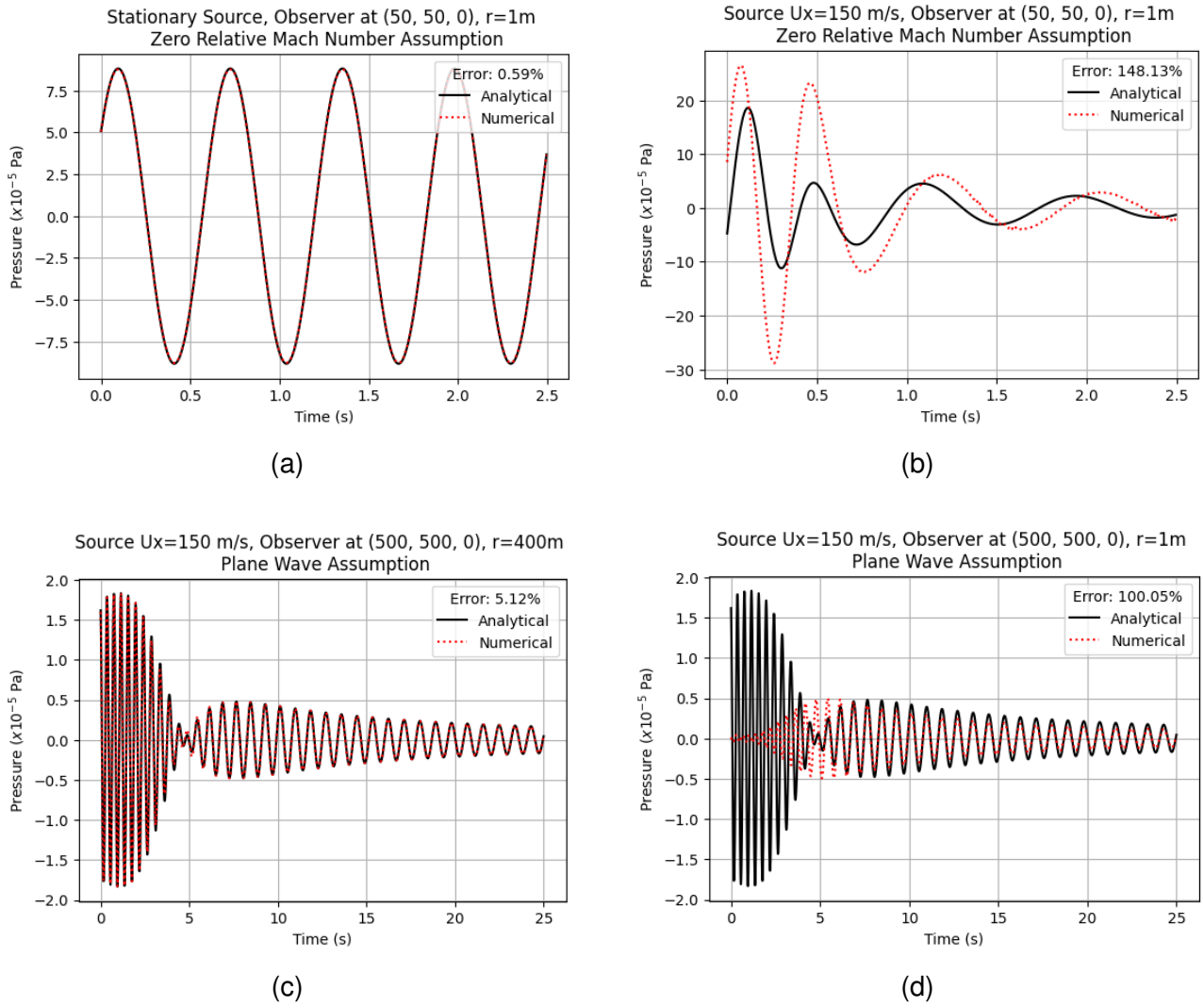


Figure 2.12: Comparison of analytical and numerical dipole. Top: zero relative Mach number case, with a) following, and b) violating the assumption. Bottom: plane wave case, with c) following, and d) violating the assumption.

For the zero relative Mach number assumption, the tests examine how relative motion affect the results. In each case, the observer is put at $[50, 50, 0]$. To examine the integration accuracy, the first case examines a stationary problem, and the second one a relative motion problem, moving the source at $150m/s$ along the x-direction. In the plane wave assumption, the tests only examine the relative motion and putting the observer at $[500, 500, 0]$, to be in the "far-field." The control surface radius is expanded to $r = 400m$ in the valid case and to $r = 1m$ in the invalid case.

For stationary source and true far-field observer (Figures 2.12a and 2.12c), the results have good agreements at $< 6\%$ error. However, introducing translation into the former and setting near-field surface to the latter violate the assumptions. For the $M_r = 0$ case, the errors are due to the non-existent relative motion in Equation 2.19; for the plane wave assumption, the control surface being in the near-field of the source. The agreement as the source moves away from the observer is due to the observer eventually being in the physical far-field.

2.6.2 Dual Monopole Validations

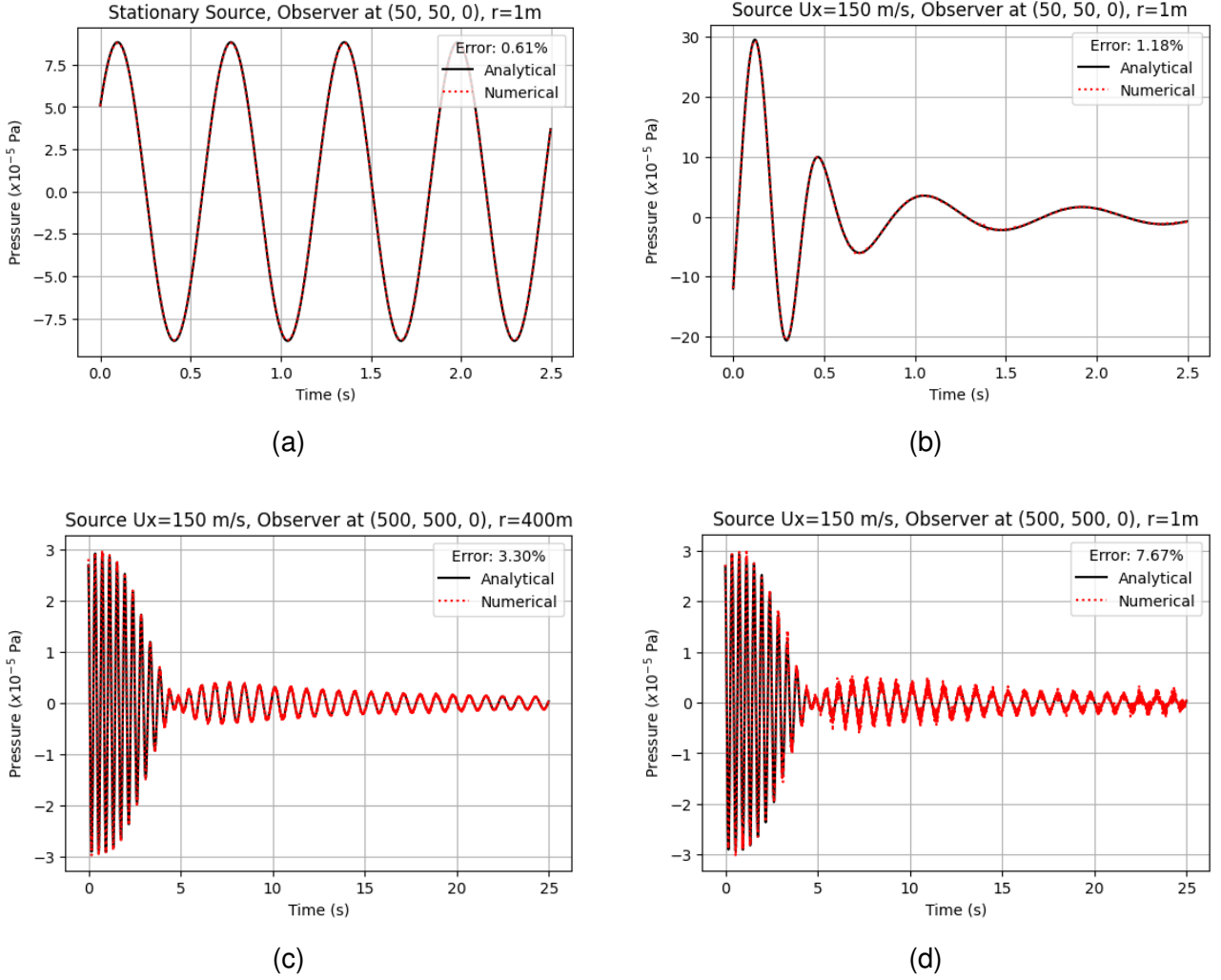


Figure 2.13: Comparison of analytical and numerical dipole validations using the dual monopole method. Top: zero relative Mach number test cases, and bottom: plane wave test cases, the numerical results for all cases agree very well with the analytical solution.

In contrast to the analytical derivations, the dual-monopole method is significantly more robust in all cases, with highest error at $< 8\%$. By superimposing two monopoles separated by $k\delta \ll 1$, the solver retains the Doppler factor, which is omitted by the zero relative motion assumption. Similarly for the bottom figures, as the monopole was validated using a control surface in the near-field, the results generally agree in the cases which the far-field assumption failed when using the same control surface. It is noted that in Figure 2.13d, the result exhibits more significant errors as the source moves away. This could be attributed to the limitations in the solver and will be discussed in the next chapter.

2.6.3 Numerical Result Comparisons for Dipole Derivations and Dual Monopoles

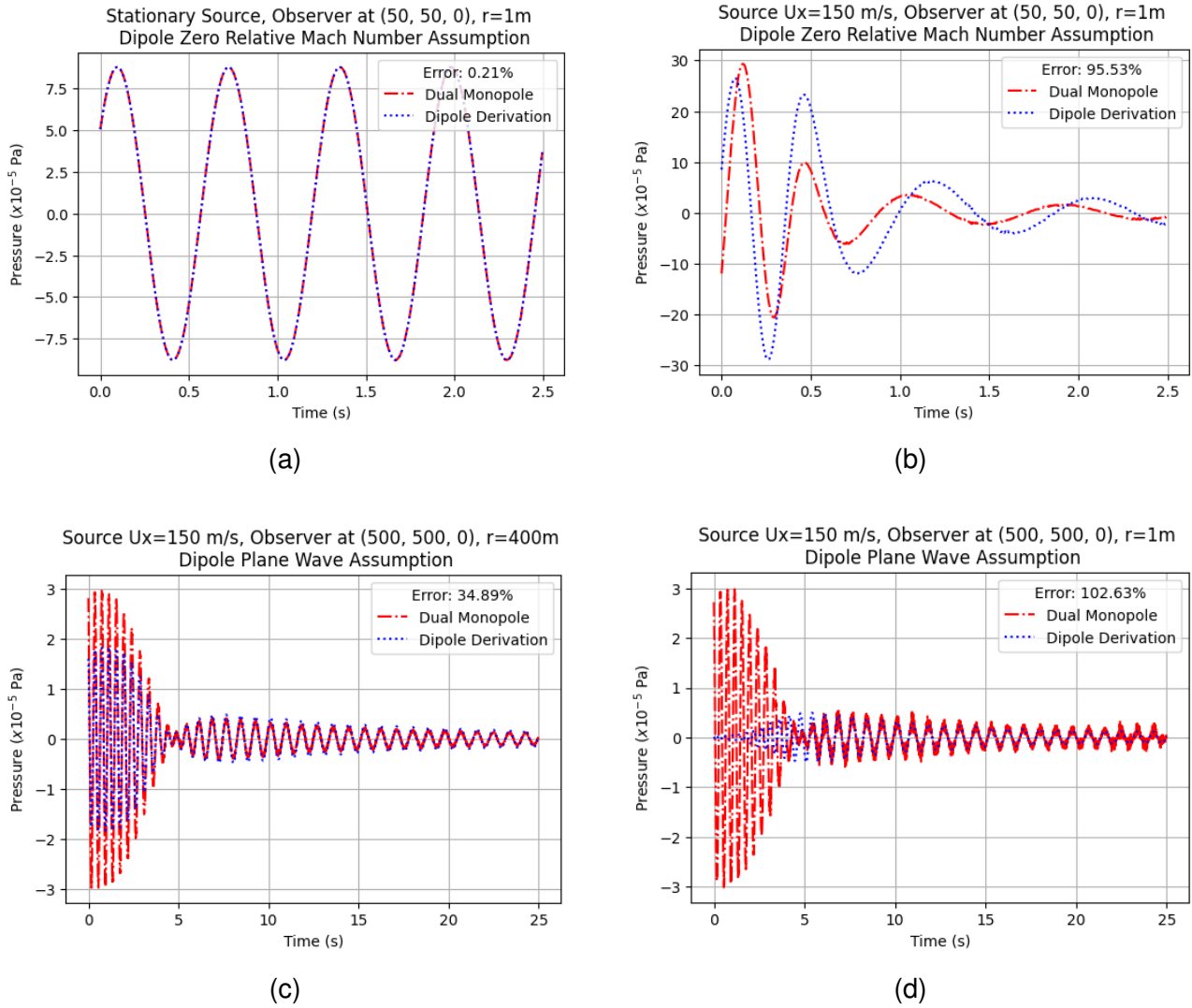


Figure 2.14: Comparison of numerical dipole validations. Top: zero relative Mach number cases, and Bottom: plane wave cases. Left column are cases where the dipole assumptions followed the analytical solutions show good agreement with dual monopole. Right column shows where the dipole assumptions are violated.

Comparing the dipole derivations to the dual monopole approach shows that there are apparent deviations in results other than the stationary case. For Figure 2.14b, it is obvious that the cause is the lack of Doppler effect in the assumption. Figure 2.14c is interesting due to a significant difference in amplitude as the sources approach the observers. Then as they move away, a good agreement in amplitude is observed. This is likely due to the imposing of the x -axis in the dipole source, because as it moves away, the y -axis contribution in the emission distance becomes insignificant.

2.6.4 Directivity

The dual monopole method allows the investigation of directivity patterns of a dipole. Two standard tests are conducted, where the dipole is stationary and another that it moves at $u_x = 10m/s$ to demonstrate the convective effects. The observers are set as a sphere located at a distance of $50m$, from the source.

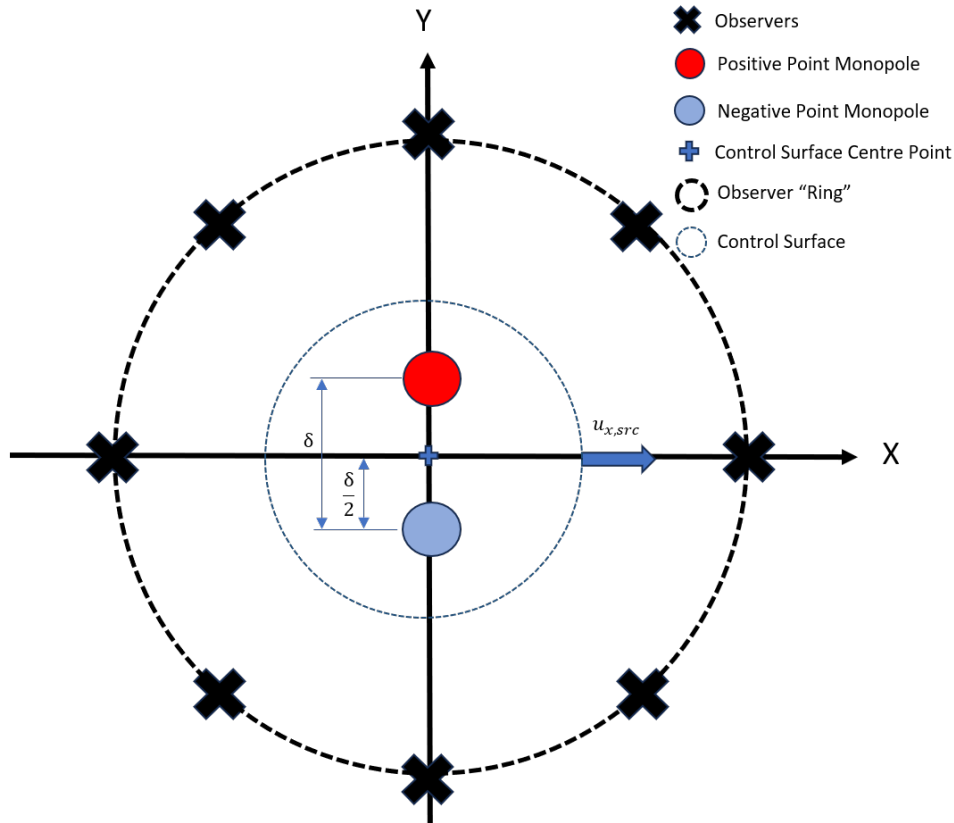


Figure 2.15: Test case for directivity

The stationary case show a standard directivity "figure-8" pattern, showing the locality of the individual monopoles. Imposing a translation to the source alters this pattern towards the x -axis as the source moves in said direction.

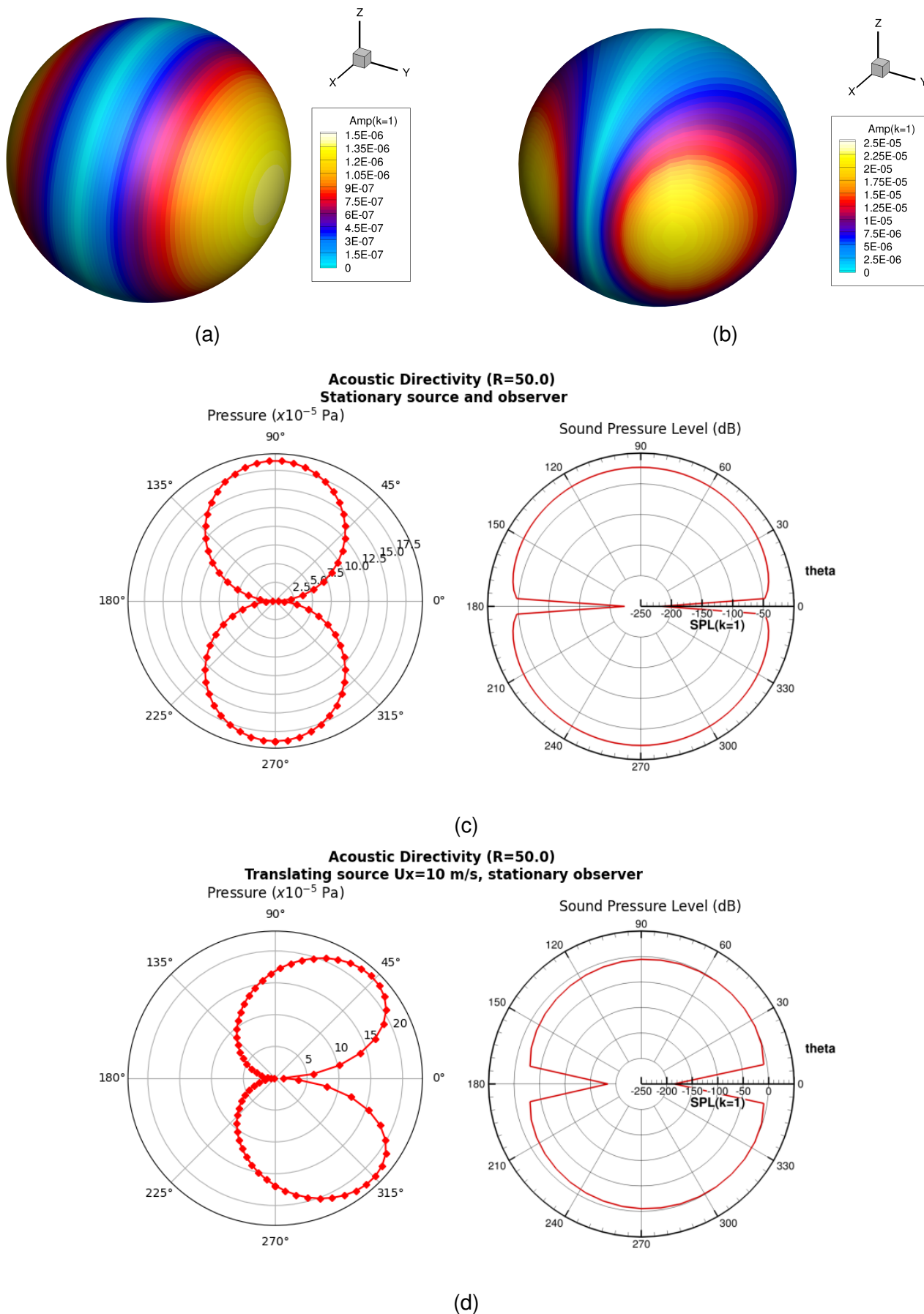


Figure 2.16: Directivity of dual monopoles, stationary case with a) surface plot and c) polar plot; and translating case with b) surface plot and d) polar plot

2.7 Concluding Summary

In the grid resolution comparisons, while the finest resolution of 40×40 has very good agreement with the analytical solutions at less than 1% error in all cases, the errors progressively increase to $\sim 4\%$ and $\sim 30\%$, for coarser resolutions. The plots show at lower resolution, the numerical results could not capture the peaks and troughs of the harmonic wave, causing dissipation. Given the acoustic wavelength is $\sim 214m$ (from the prescribed frequency and speed of sound values), the coarsest grid has an arc distance between two nodes of 1.25 meters. The errors shown in the grid sensitivities are mainly due to geometric error of the spherical domain, rather than the grids not being able to capture the wavelengths. This shows that this grid resolution is sufficient given the input and control surface radius.

The time step sensitivity study exhibits similar trends. The $dt = 0.005s$ and $dt = 0.025s$ sizes yield errors at less than 1% and 2%, respectively. However, largest time-step size of $dt = 0.125s$ (an $8Hz$ sampling frequency), is insufficient to capture the input acoustic frequency of $10rad/s$ or $\sim 1.6Hz$, as it can only sample at five times the input. This can be seen in the lags in the plots as the numerical solution attempted to keep up with the wave.

Chapter 3

Sensitivity Analysis of the FW-H Method

3.1 Sensitivity Analysis Background and Methodology

With the FW-H code validated, the next step before implementing it into CFD data is establishing its sensitivity to spatial and temporal discretisation. While the validation cases touched on these effects, this section defines the discretisation parameters that serve as the baseline for the subsequent sensitivity analysis.

3.1.1 Definition of Non-Dimensional Discretisation Parameters

XXXX MORE LITERATURE FOR THE DISCRETISATION RESOLUTIONS

In existing literatures [27, 5], discretisation is typically quantified relative to the wave properties rather than absolute physical dimensions. Spatial resolution is defined by the number of grid points per wavelength, and temporal resolution is defined by the number of time steps per wave period. Using the fundamental wave speed relationship,

$$c = \lambda f = \frac{\lambda}{T} \quad (3.1)$$

the spatial resolution parameter X is defined as the wavelength divided by the characteristic grid spacing. Similarly, the temporal resolution parameter P is defined as the wave period divided by the time-step size,

$$X = \frac{\lambda}{d} \quad \text{and} \quad P = \frac{T}{\Delta t} \quad (3.2)$$

where,

$$d = \frac{2\pi R}{N}$$

which denotes the circumferential distance between grid points in a spherical domain, as shown in Figure 1.2. These baseline definitions are sufficient for stationary sources, but require modification for high-speed kinematics, such as open rotor blades, due to Doppler effect. The relative motion between the source and the observer compresses or expands the wave received by the observer, therefore, introducing a Mach number scaling factor, $1 \pm M$ is required to both the perceived wavelength and period. Therefore, the parameters X and P must be updated to account for the Doppler-shifts, redefining X as the Doppler-shifted wavelength divided by the grid spacing, and P as the Doppler shifted period divided by the time-step size,

$$X = \frac{\lambda_s(1 - M)}{d} \quad \text{and} \quad P = \frac{T_s(1 - M)}{\Delta t} \quad (3.3)$$

It is important to note that these parameters are imperfect as they provide a nominal baseline rather than an exact local measurement. For example, when using an analytical sound source, the code uses a spherical domain constructed with a latitude-longitude grid, meaning the grid spacing are non-uniform across the longitude. Furthermore, when applying the Advanced Time formulation in CFD cases, the varying emission times project onto a non-uniform observer time. Therefore, the sensitivity tests conducted in this chapter uses the Retarded Time provide a uniform temporal discretisation to quantify the code's limits to define the source CFD's temporal discretisation.

In quantifying the code's sensitivity to the discretisation parameters, the same error quantification is used as the one used in Chapter 2, i.e., using the L2-norm to compare the instantaneous values of the analytical solution and the numerical integration results. After errors at a given P and X combinations are calculated, a 2D matrix is presented to provide the range of error at the given parameter ranges. This approach is selected as it gives a unifying approach to the error quantification as the validation methodology, as well as to provide an easily understood quantification of the code's sensitivity.

3.1.2 Background of Test Cases

The objective of this sensitivity analysis is to understand the FW-H code's behaviour across varying discretisation parameters before applying it to a real CFD case. Rotor noise is defined by two key characteristics. The first is periodicity, which is dominated by harmonics called at the Blade Passing Frequency (BPF) and its integer multiples. The second is the subsonic operating speed of the blades, which causes a Doppler shift that amplified the observed frequencies and pressure amplitudes while compressing the acoustic wavelength. The two test cases were developed to ensure the FW-H code correctly resolves these physical phenomena.

3.2 Multipole Ring Analysis

3.2.1 Background of Multipoles Analysis

To address the multi-harmonic characteristic of the propeller noise, a stationary ring of monopoles, referred to as the multipole ring case, is used. This setup replicates the thickness (monopole) noise data first established by Farassat [28] for a three bladed propeller, which remains a benchmark for research in aeroacoustics [29, 30]. Instead of modelling the blades, discrete point sources form a ring, with a radius equal to the reference propeller. This approach represents the "blade passing points," where the discrete sources represent thickness of the rotor blade tips passing through the ring's azimuthal positions.

3.2.2 Mimicking the Noise Signature of a Propeller using Analytical Multipoles

3.2.3 Sensitivity of the FW-H Method in Multipole Ring Prediction

3.3 Sensitivity Analysis of Rotating Monopoles

3.3.1 Background of Rotating Monopole Analysis

The second test approaches the second characteristic of the rotor noise by modelling a sound source in subsonic motion. Following the work by Ffowcs Williams [31], a single monopole point rotates at specific Mach numbers while emitting pressure waves. The radiation frequency (f_{source}) is defined as an integer multiple (n) of the rotational frequency (f_{rot}), such that $f_{source} = n f_{rot}$.

3.4 Concluding Summary

Chapter 4

Application of the FW-H Method on Open Rotor Engine CFD Model

Chapter 5

Conclusions

A Ffowcs Williams-Hawkings (FW-H) acoustic code has been analytically validated. The numerical integration is accurate, matching analytical derivations for stationary and moving monopole sources with less than 1% error under sufficient grid and temporal resolution.

For dipole sources, the explicit analytical derivations were shown to break down when subjected to non-zero relative motion or near-field control surfaces. To bypass these limitations, a dual-monopole approach was implemented, proving to be a good test case to demonstrate dipoles. While the numerical integration is mathematically sound, further refinement of the dipole analytical validation method is required. Once complete, the code will be integrated with URANS CFD for open rotor noise prediction.

Chapter 6

Future Work

With the FW-H solver validated with some limitations, the project moves to the hybrid analytical-computational framework. The next part of the work include:

- Validation refinements for dipoles (e.g., using non-stationary monopole as the source term, dynamic vector calculations) to reduce errors.
- Updating the numerical FW-H integration code to use double-precision floating-point arithmetic.
- Development and validation of a baseline URANS-FW-H hybrid model.
- Using the validated URANS-FW-H approach to predict open rotor noise.

Review of Generative AI Use

Generative AI (Google Gemini 3 Pro) has been used in augmenting the research and writing of this report. The usage of which are in helping with LaTeX formatting, formalising some English language uses, mathematical derivations, and checking for programming errors or bugs. The following are prompt examples used on the above main usages:

- "Help me format this LaTeX document using the structure abstract, table of contents, nomenclature, intro, lit review, derivations, validations, discussion, conclusions, future work, appendices. The word count should only include the main content (from intro to future work).
- "Help me rephrase these two sentences and bridge between them"
- "In deriving the velocity equation for the dipole, as the explicit integral of the convolution between the right hand side term and Green's function is unviable, what would be the implications of assuming that r is a constant value, such that some integrations of the $F/(1-M_r)$ term can be done? If this is solved in the retarded time, the denominator is cancelled, so I can integrate F directly, correct?"
- "As the dual monopole method requires introducing an additional point source, which the code had not considered initially, help me to put an additional point monopole in the `sample_properties` subroutine in Fortran. As per the derivations, I need to separate the two by a distance δ , which I think can just be calculated as $0.01/k$ so it dynamically changes if any parameters are changed in the future.

The result of the first two examples being in the formatting of this report and some of the sentences within. None of the result sentences are directly copied from the tool, and instead are rephrased in the author's own words to reflect the author's understanding of the topic. The third prompt helped the author in deriving the mathematical parts presented in this report; of which the results are checked and verified via textbooks and during meetings with supervisors. The fourth prompt example returned Fortran codes that are used for the analytical solutions. These solutions tend to come with bugs which the author fixed independently.

In conclusion, all text-based outputs given by the Generative AI were checked, reworded, and edited to be based on the author's understanding of the topic and to avoid plagiarism. Outputs that are technical are verified by textbooks and discussions with supervisors to

ensure the accuracy of the results and limitations of the assumptions.

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Appendix A

Research Project Gantt Chart

The following is the Gantt Chart of the current project up to its completion. Some adjustments had to be made along the duration of the project's execution such as CFD-related parts, however, the overall project objectives and milestones remain the same.

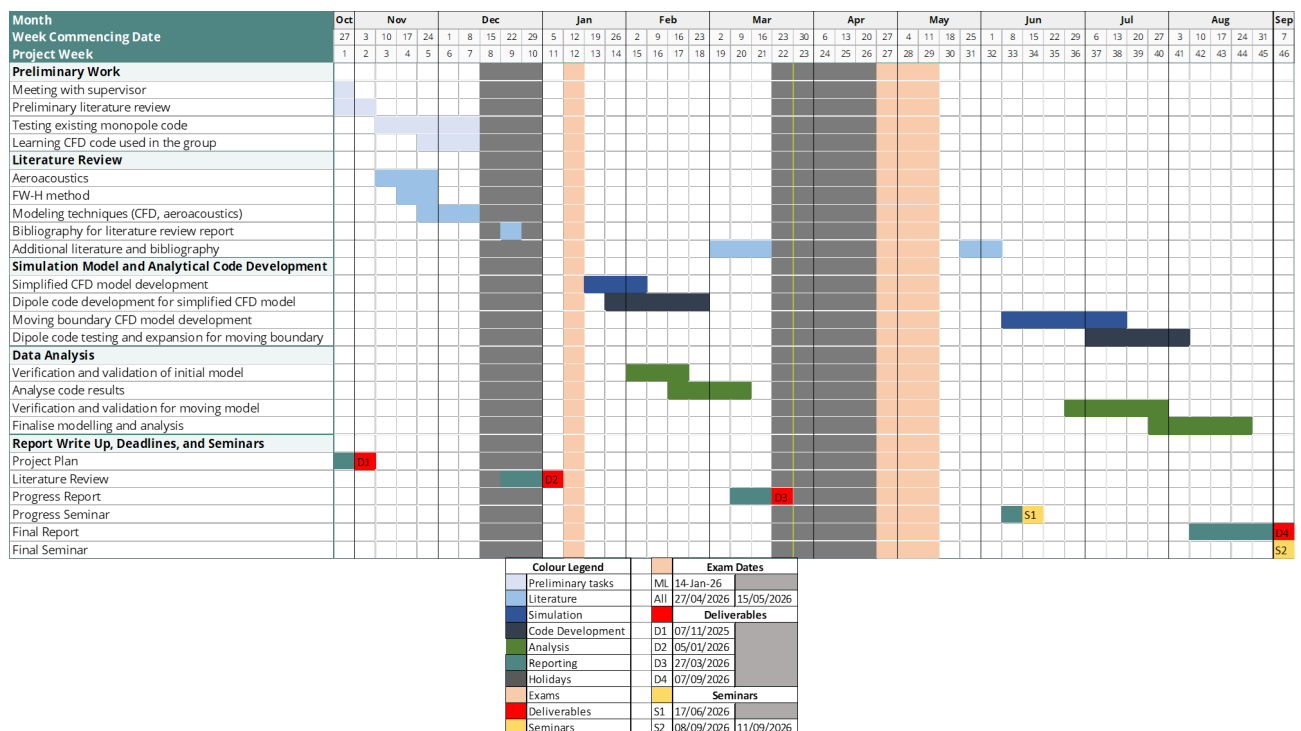


Figure A.1: Gantt Chart of the research project

Appendix B

Variable Derivatives

To derive the analytical point sources, a few definition for the derivatives are established.

$$\left. \frac{\partial r}{\partial \tau} \right|_x = -v_r \quad (\text{B.1})$$

$$\left. \frac{\partial M_r}{\partial \tau} \right|_x = \left(\frac{\partial M}{\partial \tau} \right)_r + \frac{c(M_r^2 - |M|^2)}{r} \quad (\text{B.2})$$

$$\left. \frac{\partial |1 - M_r|}{\partial \tau} \right|_x = \frac{(1 - M_r)}{|1 - M_r|} \left(- \left(\frac{\partial M}{\partial \tau} \right)_r + \frac{c(|M|^2 - M_r^2)}{r} \right) \quad (\text{B.3})$$

$$\left. \frac{\partial [Q(\tau)]_{\tau^*}}{\partial \tau} \right|_x = \left[\frac{1}{1 - M_r} \frac{\partial Q(\tau)}{\partial \tau} \right]_{x|\tau^*} \quad (\text{B.4})$$

$$-\frac{\partial}{\partial x_i} \Big|_t \int_S [Q_i]_{\tau^*} dS = \frac{\partial}{\partial t} \Big|_x \int_S \left[\frac{Q_i r_i}{cr} \right]_{\tau^*} dS + \int_S \left[\frac{Q_i r_i}{r^2} \right]_{\tau^*} dS \quad (\text{B.5})$$

Appendix C

Green's Function for Wave Equations

Given a wave equation,

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \psi(\hat{r}, t) = -f(\hat{r}, t)$$

Green's function of the above equation,

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) G(\hat{r}, t | \hat{r}', t') = -\delta(\hat{r}' - \hat{r})\delta(t - t')$$

Let $\hat{r}' = 0$ and $t' = 0$. Solving for $\nabla^2 G$

$$\begin{aligned} \nabla^2 G(\hat{r}, t) &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) G \\ &= \frac{1}{r} \frac{\partial G}{\partial r} + \frac{1}{r} \left(\frac{\partial G}{\partial r} + r \frac{\partial^2 G}{\partial r^2} \right) \\ &= \frac{1}{r} \frac{\partial^2}{\partial r^2} (rG). \end{aligned}$$

Substituting back to the wave form,

$$\begin{aligned} \nabla^2 G - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} G &= -\delta(r)\delta(t) \\ \frac{1}{r} \frac{\partial^2}{\partial r^2} (rG) - \frac{1}{c^2} \frac{1}{r} \frac{\partial^2}{\partial t^2} (rG) &= -\delta(r)\delta(t) \end{aligned}$$

Factoring the differential terms, it is found that if $r > 0$, then the left-hand side term is equal to zero,

$$\left[\left(\frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial}{\partial r} \right) \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial r} \right) \right] (rG) = 0 \quad (\text{C.1})$$

Then, let $u = ct - r$ and $v = ct + r$, it will be found that,

$$\frac{\partial}{\partial u} = \frac{1}{2} \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial r} \right) \quad \frac{\partial}{\partial v} = \frac{1}{2} \left(\frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial}{\partial r} \right)$$

Given that $\partial^2(rG)/(\partial u \partial v) = 0$, if $rG = g(v) = g_+(ct + r)$ and $rG = g(u) = g_-(ct - r)$, then,

$$G = \frac{1}{r} g_{\pm}(ct \pm r)$$

$$G(\hat{r}, t | \hat{r}', t') = \frac{1}{\hat{r} - \hat{r}'} g_{\pm}(c(t - t') \pm |\hat{r} - \hat{r}'|)$$

Taking $\nabla^2 G$ using this definition,

$$\begin{aligned} \nabla^2 G &= \nabla^2 \left(\frac{1}{r} g_{\pm}(ct \pm r) \right) = \nabla^2 \left(\frac{1}{r} \nabla g_{\pm} + g_{\pm} \nabla \left(\frac{1}{r} \right) \right) \\ &= -\frac{2}{r^2} \frac{\partial g}{\partial r} + \frac{2}{r^2} \frac{\partial g}{\partial r} + \frac{1}{r} \frac{\partial^2 g}{\partial r^2} - 4\pi \delta(r) g \end{aligned}$$

The first two terms of the right-hand side cancel, and moving back to the wave equation format,

$$\nabla^2 G - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} G = -\frac{1}{r} \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial r^2} \right) g_{\pm} - 4\pi \delta(r) g_{\pm} \quad (\text{C.2})$$

As described in equation C.1, the first term of the right-hand side cancels. Going back to the wave equation,

$$\square^2 \psi(\hat{r}, t) = -\delta(\hat{r}) \delta(t) = -4\pi \delta(\hat{r}) g_{\pm}$$

Therefore,

$$\therefore G_{\pm} = \frac{\delta(ct \pm r)}{4\pi r} \quad (\text{C.3})$$

Appendix D

Replacing $\partial/\partial x_i$ Terms

As the definition of dipoles are that of the spatial derivative of the forcing term convolved with the Green's function in 3D free-space, it is necessary to replace the spatial derivative terms to that of the temporal one to derive using the definitions described in Appendix B. As the goal is to transform $\mathbf{x} \rightarrow \mathbf{X}$ and $t \rightarrow \tau$, and the following are true: $\mathbf{x} = \mathbf{X}$ and $\tau = t - \frac{r}{c}$, then

$$\begin{aligned}\left. \frac{\partial}{\partial x_i} \right|_t &= \frac{\partial X_j}{\partial x_i} \frac{\partial}{\partial X_j} + \frac{\partial \tau}{\partial x_i} \frac{\partial}{\partial \tau} = \delta_{ij} \frac{\partial}{\partial X_j} - \frac{r_i}{cr(1 - M_r)} \frac{\partial}{\partial \tau} \\ \frac{\partial}{\partial x_i} &= \frac{\partial}{\partial x_i} - \frac{r_i}{cr(1 - M_r)} \frac{\partial}{\partial \tau}\end{aligned}\tag{D.1}$$

Similarly for the time derivative,

$$\left. \frac{\partial}{\partial t} \right|_{\mathbf{x}} = \frac{\partial X_j}{\partial t} \frac{\partial}{\partial X_j} + \frac{\partial \tau}{\partial t} \frac{\partial}{\partial \tau}$$

Assuming no mean flow, the first term of the right hand side, $\partial X_i/\partial t$ is equal to zero, leaving,

$$\left. \frac{\partial}{\partial t} \right|_{\mathbf{x}} = \frac{1}{1 - M_r} \left. \frac{\partial}{\partial \tau} \right|_{\mathbf{x}}\tag{D.2}$$

Appendix E

Monopole Velocity Derivations

Using the momentum equation, assuming the flow has no mean flow and is inviscid, substituting the monopole definition into the pressure term, the second terms of both left- and right-hand are zero. Defining the pressure gradient by defining the pressure as monopole pressure definition,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial}{\partial x_i} \frac{\partial}{\partial t} \left[\frac{Q(\tau)}{4\pi r(1 - M_r)} \right]_{\tau*}$$

and cancelling the unsteady term from both sides gives the following for the momentum equation,

$$u_i = \frac{1}{\rho_0} \frac{\partial}{\partial x_i} \left[\frac{Q(\tau)}{4\pi r(1 - M_r)} \right]_{\tau*} \quad (\text{E.1})$$

Convolving with Green's function in free 3D-space is done to replace the spatial derivative into a temporal derivative,

$$p'(\mathbf{x}, t) = \frac{\partial}{\partial x_i} (Q_i \star G) = -\frac{\partial}{\partial t} \left(Q_i \star \frac{x_i G}{c|\mathbf{x}|} \right) - Q_i \star \frac{x_i G}{|\mathbf{x}|^2}$$

which has the same form as the integral shown in Appendix B, Equation B.5. Substituting the monopole source terms into Q_i , and converting to retarded time using the equation found in Appendix D Equation D.2, gives,

$$\frac{\partial}{\partial x_i} \left[\frac{Q(\tau)}{4\pi r(1 - M_r)} \right]_{\tau*} = \frac{1}{1 - M_r} \frac{\partial}{\partial \tau} \left[\frac{Q(\tau)r_i}{4\pi c r^2(1 - M_r)^2} \right]_{\tau*} + \left[\frac{Q(\tau)r_i}{4\pi r^3(1 - M_r)} \right]_{\tau*}$$

which can then be solved analytically to give the solution for velocity,

$$u_i = \frac{1}{4\pi\rho_0} \left[\frac{(\partial Q(\tau)/\partial\tau)r_i}{cr^2(1-M_r)^2} + \frac{Q(\tau)r_i}{r^3(1-M_r)^3} + \left(\frac{dM}{d\tau} \right) \frac{Q(\tau)r_i}{cr^2(1-M_r)^3} \right. \\ \left. - \frac{Q(\tau)M^2r_i}{r^3(1-M_r)^3} - \frac{Q(\tau)M_i}{r^2(1-M_r)^2} \right]_{\tau^*} \quad (\text{E.2})$$

Appendix F

Dipole Velocity: Zero Relative Mach Number Assumption Derivations

Defining pressure gradient from the momentum equation and dipole definition,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial^2}{\partial x_i \partial x_j} \left[\frac{f_i(\tau)}{4\pi r(1 - M_r)} \right]_{\tau^*}$$

Going back to momentum equation, the velocity is,

$$u_i = \frac{1}{4\pi\rho_0} \frac{\partial^2}{\partial x_i \partial x_j} \int_{-\infty}^t \frac{f_i(\tau)}{r(1 - M_r)} dt$$

assuming $dr/dt = 0$ and $M_r = 0$, letting $F_i(\tau) = \int f_i(\tau) d\tau \implies f_i(\tau) = \dot{F}_i(\tau)$, and changing to integrating with respect to τ , this simplifies the integrand to,

$$u_i = \frac{1}{4\pi r \rho_0} \frac{\partial^2}{\partial x_i \partial x_j} \int_{-\infty}^t f_i(\tau) d\tau$$

Taking the integral,

$$u_i = \frac{1}{4\pi r \rho_0} \frac{\partial^2}{\partial x_i \partial x_j} \int_{-\infty}^t f_i(\tau) d\tau = \frac{1}{4\pi r \rho_0} \frac{\partial^2}{\partial x_i \partial x_j} [F_i(\tau - r/c)]$$

Hence, the analytical solution for velocity that is valid for cases of zero relative Mach number between the source and observer for near- and far-field,

$$u_i = \frac{1}{4\pi\rho_0} \left(\frac{F_j(3n_i n_j - \delta_{ij})}{r^3} + \frac{f_j(3n_i n_j - \delta_{ij})}{cr^2} + \frac{\dot{f}_j(n_i n_j)}{c^2 r} \right) \quad (\text{F.1})$$

Appendix G

Dipole Velocity: Plane Wave Assumption Derivations

Given the plane wave,

$$p'(\mathbf{x}, t) = \frac{g(t - r/c)}{r} \quad (\text{G.1})$$

Taking the pressure gradient,

$$\frac{\partial p'}{\partial x_i} = \left[\frac{\partial}{\partial x_i} \frac{1}{r} g \right] + \frac{1}{r} \left[\frac{\partial}{\partial x_i} g \right]$$

where the first and second terms correspond to the near- and far-field, respectively. The plane wave is valid only in the far-field, and taking the time derivative yields,

$$\frac{\partial p'}{\partial t} = \left[\frac{\partial}{\partial t} \frac{g}{r} \right] = \frac{1}{r} \frac{\partial}{\partial t} g(\tau) - g(\tau) \frac{v_r}{r^2}$$

In the far-field, $\frac{1}{r} \frac{\partial}{\partial t} g(\tau) \gg g(\tau) \frac{v_r}{r^2}$, simplifying the expression to,

$$\frac{\partial p'}{\partial t} \approx \frac{1}{r} \frac{\partial}{\partial t} g(\tau)$$

which gives the following expression,

$$\frac{\partial p'}{\partial x_i} = -\frac{n_i}{c} \frac{\partial p'}{\partial t}$$

substituting into the momentum equation provides the velocity equation for the plane wave,

$$u_i = \frac{n_i}{\rho_0 c} p'(\mathbf{x}, t) \quad (\text{G.2})$$

